claude-windows / 96963dab-d38f-488c-af44-530ddbeefed1

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer\96963dab-d38f-488c-af44-530ddbeefed1\subagents\workflows\wf_00bebd3c-82f\agent-a0f2bcb0972a20201.jsonl`
- SHA-256: `1ee9a680a157e6ced0cb381313f0107d131305669c6a385ff652dd3e3849572e`
- Source modified: `2026-07-08T05:51:52+00:00`
- Imported at: `2026-07-08T15:59:27+00:00`
- project: `wf_00bebd3c-82f`
- session_id: `96963dab-d38f-488c-af44-530ddbeefed1`

Transcript

1. user (2026-07-08T05:44:49.641Z)

```
You are mapping a subsystem of the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (Windows checkout; read-only ? do NOT edit anything).
Context: GitHub issue #524 wants to (a) move validation onto the L4 Cloud Run Job worker via a config-driven phase_placement knob, (b) drop a redundant re-hash in /api/process by trusting the upload's streamed video_id, and (c) design a topology where upload->stitch runs on the L4 with async copy-to-bucket. You are producing the precise code map a designer/implementer needs.
Cite exact file paths and line numbers. Read the actual code, not just docs. Follow imports where the flow crosses files.
Subsystem: DOCS + CONVENTIONS. Read docs/COMPUTE_DECOUPLED_SERVING/ (the main design doc + runlogs dir listing), docs/PROJECT_DOC_TEMPLATE.md, docs/DOC_STATUS.md (lifecycle rules), and the ADRs for 012 (Cloud Run+GPU serving), 013 (GCP-only compute), 014 (compute-decoupled ratification) in docs/archives/decisions/DECISIONS.md. Summarize: (1) the structure/status-header conventions a NEW design doc under docs/COMPUTE_DECOUPLED_SERVING/ must follow, (2) the M0-M9 milestone framing of COMPUTE_DECOUPLED_SERVING and where a "phase placement / L4-primary" design fits, (3) any prior cost analysis (Cloud Run GPU pricing, per-run cost numbers) already in the docs that the new doc should reuse or update, (4) exact constraints from ADR-012/013/014 that bound the topology options (e.g. GCP-only, Cloud Run chosen over GCE ? why, and is that ADR binding for serving topology?).
```

2. user (2026-07-08T05:45:02.508Z)

```
architecture.svg
README.md
runlogs
TESTING_STRATEGY.md
```

3. user (2026-07-08T05:45:09.630Z)

```
1       # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2
3       > **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** · **Owner:** Avi · **Created:** 2026-06-21 · **Last updated:** 2026-06-21
4       > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/past_projects/` when DONE)
5       > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state` + `docs/NEXT_STEPS.md` once work starts.
6       > **Relation to existing docs:** the productionization successor to the archived `DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from [PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
7       > **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed **compute-decoupled serving** because **truly-local is a co-equal, first-class profile**, not an afterthought.
```

> **Honesty note:** claims tagged `[verified]` (checked against code by the design workflow `wf_56274ff0`), `[design]` (proposed), `[researched]`.

> **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13, a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user *confirms* (D15).

0. TL;DR
The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB / mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local** (local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive / bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus **cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the **parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.

1. Problem & goal
- **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4) and is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero + cost guard)** become this project. The owner needs **both** a hosted service *and* a truly-local mode, with the **core (detect+stitch) unaffected** by which one runs.
- **The two user-facing outcomes** (owner, 2026-06-21):
  - **(a) Hosted service** ? operate on small **local uploads (`<2GB`)** *and* load from **gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching** (stitching is compute-intensive ? run it where its cost is accounted for).
  - **(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local GPU**, run inference **and** stitching **completely locally**.
- **"Good" looks like:** the same `video ? rallies ? reel` core gives identical windows + stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup choice**, never a code branch in the core.

2. Decisions locked

#	Decision	Source	Implication
D1	The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile branch inside it**.	design `[verified]`	All profile difference lives in config + the two registries.
D2	**Two registries:** `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`.	ADR-014	Realizes PLATFORM_ARCHITECTURE's `local|hosted|byo_worker`.
D3	**Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred.	ADR-014	Enum/seam reserved so BYO isn't precluded.
D4	In **cloud-serving, STITCH runs on Cloud Run** (co-located with detect) so CPU/wall + egress are metered into the per-job cost ledger.	owner (a)	Stitch-on-laptop would hide real cost.
D5	**Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory); **detect** is byte-exact only same-GPU.	deploy report 2026-06-20	Cross-GPU asserted at the rally/window level, never CSV bytes.
D6	**Default-OFF, smallest-safe-first, one PR per milestone**; any core-caller change (e.g. configurable output base) ships behind a flag + a Launch Report.	[[launch-report-convention]]	Zero-regression discipline.
D7	**Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that ~$500/mo idle would break per-video billing).	owner 2026-06-21	M6/M7. Revisit a warm lane only on a UX complaint.
D8	**Pricebook = a versioned DB table**; cost computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore		

market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |
36 | D9 | **Edge auth = Cloudflare Access** (reuse the site's existing CF). Lock the Cloud Run **ingress to the CF proxy** + **verify the `Cf-Access` JWT signature** (so a direct hit to the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity system. |
37 | D10 | **Free-tier "data-for-reels" trade:** free reels in exchange for **training rights** (uploaded footage + derived trajectories/labels); **paid tier = no-train** (privacy is the paid feature). **Carve-outs:** train ONLY on attested **ADULT** footage (minor footage ? reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on **DERIVED** data + **auto-delete raw**; **ToS counsel-reviewed**; live training-on-uploads **gated to M9** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel. |
38 | D11 | **Quality floor: Gemini handoff ON** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |
39 | D12 | **Data-flywheel harness is IN SCOPE (owner add, Q5):** capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones to the golden TRAINING set** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new **M11**; ties D10 (consent) + `data_pipeline/` + the eval/experiment harness. |
40 | D13 | **Profile is user-switchable ANYTIME ? a headline FEATURE.** The same user runs **local today, cloud tomorrow, back to local** ? it's just a config/startup choice; the pure core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the **rally/window** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is *between* jobs. |
41 | D14 | **NORTHSTAR ? operate toward this:** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. | owner 2026-06-21 | Implies a **`gdrive-api` (OAuth) StorageBackend** (read source + write reel back) ? distinct from the local **mount** backend; **resolves §8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude it (the `StorageBackend` ABC accommodates it). |
42 | D15 | **Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.** A GPU owner may still choose cloud; **cloud (= spend) is never entered silently**; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends §4.3 (auto-detect = a proposed default, not an auto-decision). |
43 | D16 | **M6/M7 implementation SHAPE + the cheap §8 picks (owner 2026-06-21):** (a) **Cloud Run JOBS** (not Services-with-GPU) for the detect+stitch unit, **bucket-poll for completion** (reuses `JobWaiting`/`claim_due_job` verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) **stitch ENCODE = libx264** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? **NVDEC for DECODE only** (a speed lever that never touches output bytes); **NVENC encode is DEFERRED behind a config knob**, revisit only if M7 cost data justifies relaxing parity (§8 Q8); (c) **input cap = hard-reject `<2GB`** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? §8 Q9). | owner 2026-06-21 | M6/M7. Resolves §8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |
44 | D17 | **Real cloud verification BUDGET = $100 / ?10k cap (owner 2026-06-21):** the owner authorized actual GCP spend for due-diligence + production-grade testing/verification of M5?M7 (real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), **provided the cap is respected** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and **DELETE every resource after** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |
45
46 ## 3. Foundation ? evolution / ADR lineage ? *trace the idea end-to-end*
47 This project is the latest step in a long decision chain; each ADR contributed a piece and a **subtlety that carries forward**:
48
49 | ADR / doc | Contributed | Subtlety carried into this project |
50 |---|---|---|
51 | **ADR-005** | Permissive licensing (MIT/Apache-2.0/BSD only) | The engine + any served

model + the **container image** (ffmpeg, fonts, weights) stay commercial-clean. |
ADR-008	Owned-model topology; **train==serve featurizer single-sourcing**	The "core unaffected across profiles" is the same parity discipline, extended from train/serve to **local/cloud**.
ADR-009	KhelSutra Guru; **local-first delivery**	The **truly-local profile** is the direct continuation ? privacy/offline self-host stays first-class.
ADR-010	DECOUPLED: DO?GCP pivot	The cloud-compute origin (GPU + co-located bucket).
ADR-011	DECOUPLED scope = M0?M2+M3.5; **deferred M3 (serving) + M4 (billing/cost-guard)**; overrode the *"rent nothing / not a service"* posture	**This project realizes M3/M4.** Carries §06's owner-gated gates: **DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3**.
ADR-012	GPU-native; TPU deferred; **Cloud Run+GPU scale-to-zero = serving model**; **NVDEC** named as the decode lever	The serving substrate + the 7-step seed scope this expands.
ADR-013	Drop DO; **GCP sole compute stack**; $200 DO credits ? non-compute only	The provider for cloud-serving; DO Spaces remains only a possible S3 *storage* target.
? ADR-014 (new)	**Compute-decoupled serving:** one pure core, profile-selected compute+storage; stitch-where-metered	This doc. **Refines** ADR-011/012, does **not** supersede them.
PLATFORM_ARCHITECTURE.md	The `ComputeBackend`/`StorageBackend` registries	The abstraction this realizes in code.
VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md	L0 fully-local tier / `LocalDesktop` backend	The truly-local profile = these, productionized.
07-COMPUTE-ABSTRACTION (archived)	`DeviceContext`/`DeviceProfile` (designed)	WASB hardcodes `device='cuda'` with **no CPU fallback** ? a portability gap M4 closes.
DEPLOY_REPORT_GCP_2026-06-20 (archived)	First real L4 run; **cross-GPU sub-pixel parity finding**	Why D5 (parity at window level, not CSV bytes).

4. Design / architecture
See [`architecture.svg`](architecture.svg) for the picture. **The core never changes; the box around it does.**

4.1 The core contract (must stay byte/behaviour-identical) `[verified]`
1. **DETECT (GPU):** `DetectorRunner.run_predict(video_path, output_dir, video_id) ? CSV[Frame,Visibility,X,Y]` (`backend/pipeline/detectors/base.py:164`). Three interchangeable impls ? `WasbRunner` (WSL), `NativeWasbRunner` (Linux GPU), `StubDetectorRunner` (CPU mock) ? emit the **identical CSV schema**. The trajectory CSV is the stable artifact boundary.
2. **WINDOW (CPU pure fn):** `trajectory_to_action_windows(points, fps, frame_width, ?) ? windows` (`tracknet_runner.py:281`). A shared `@staticmethod`, zero GPU/IO/randomness ? same trajectory + config = byte-identical candidate windows. Used by every runner *and* the eval/regression stack verbatim.
3. **STITCH (CPU/ffmpeg):** `VideoCompiler.compile_highlights(raw_video, segments, out_name) ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffprobe ? watermark ? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.

Non-goal (the trap to avoid): *no* code branch inside detect/window/stitch keyed on profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-identical to CONTROL).

4.2 The deployment profiles
Profile	Compute	Storage	Orchestration	When
truly-local	`compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto` (WSL); stitch **in-process**	`local` (or `gdrive` = mounted Drive)	`python -m backend.main` or `worker infer` CLI; existing async job worker	Self-host / offline / privacy / dev-on-a-GPU-box ? owner (b)
cloud-serving	`compute_target=cloud_run`; `detector_impl=native` (L4, cuda); **detect AND stitch on the same Cloud Run job**	`gcs` (or gdrive mount / assembled upload); per-video workspace; reels via signed URL	Cloud Run control plane enqueues ? Cloud Run GPU job pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero	Hosted SaaS ? owner (a)
cpu-mock	`compute_target=cpu_mock`; `detector_impl=stub` (synth/replay)	`local`/in-memory fakes	`run_all_tests.py` / pytest	CI, regression, profile-parity, GPU-free dev

| *byo_worker* (deferred) | tenant GPU via outbound pull agent | tenant's own bucket (signed URLs) | long-poll pull; same job table scoped by tenant | Privacy-sensitive enterprise ? **DEFERRED**, enum/seam reserved |

4.3 Profile selection (the "startup/setup config") `[design]`
ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-detect in `load_config`:
- `deployment.profile ? {truly-local | cloud-serving | cpu-mock}` (empty = today's manual knobs, **fully backward-compatible**).
- Selecting a profile applies a **coherent preset bundle** of *existing* knobs that today drift independently: **INPUT** (new `input_backend`/`input_root`, parallel to the output-side `StorageConfig`), **COMPUTE** (new `compute_target` enum locking `detector_impl` + `skip_ai_handoff` + workspace strategy), **STORAGE** (`storage.backend`/`workspace_root`/`single_folder_per_video`, all already exist).
- **Precedence:** built-in defaults ? profile preset ? `config.json` ? env (`RALLY_DETECTOR_IMPL`, `WASB_*`, `DEPLOYMENT_PROFILE`, ?) ? worker `--set k=v`. Every knob stays individually overridable.
- **Env auto-detect SUGGESTS, the user CONFIRMS (D15 ? no surprise execution):** the env gives a *proposed* default (`K_SERVICE` ? cloud-serving; `CI=true` ? cpu-mock; else `torch.cuda.is_available()` ? truly-local), but the user **explicitly confirms or overrides** it ? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast** (cloud-serving without GCS creds/GPU ? loud error before any job).
- The worker (`cmd_infer`/`cmd_train`) is **already profile-agnostic** (reads `config.storage`, accepts `--config`/`--set`) ? no worker rewrite.

4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In **cloud-serving, stitch is co-located in the Cloud Run job** so `gpu_active_seconds` (detect) + `wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ? the per-job cost ledger. Running stitch on a laptop would leave that cost **invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get bursty ? the job table + `claim_due_job` already supports both.)

4.5 Reuse map ? most of the seams already exist `[verified]`
Exists: the 3-stage core; `_resolve_runner` (the single compute seam); `StubDetectorRunner` (`replay_csv` = byte-exact $0 anchor); `StorageBackend` ABC + 4 backends + `StorageRef.parse_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control plane (`jobs` table, `claim_due_job` CAS); `ExecutionPolicy WAIT_AND_RESUME`/`JobWaiting` (carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip_ai_handoff` (LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial:** per-video workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user_id` columns exist). **Net-new:** the `deployment` section; `input_backend`; configurable output base (kill hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-auth header extraction; startup env detection.

4.6 The data-flywheel harness (D12 / M11) `[design]`
Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline additionally:
1. **Captures** the source upload + the Gemini reel + the rally output (intervals/report) into the per-video workspace (`workspaces/<video_id>/`), instead of the no-consent auto-delete path.
2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw video kept only for the training pool, not the no-consent path).
3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini (`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").
4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING rows** (human-reviewed labels via the annotation flow ? `data_pipeline/`), growing the corpus ? the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.

This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the PERSONALIZATION contribution model): every consented free reel makes the owned model better. Reuses `data_pipeline/`, `backend/eval/`, and the M9 consent gate; **adults-only**, raw-retention gated by the D10 consent.

6. Honest limits ? what this does NOT do / prove
- A green CI suite proves **plumbing + determinism (window+stitch) + profile-parity** for $0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-side, real-video).
- **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a CSV-byte claim.
- The cloud `0.1.0-l4` weights are **n=2 plumbing-grade** ? cloud-serving ships with **Gemini handoff ON** until the owned model clears a floor (open Q).
- This does not resolve **WASB-C3** (sharing a trained model) or the **DPA/consent** gate ? both remain owner/counsel-gated before public, non-owner serving.

7. Deliverables & progress tracker ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off `origin/master`, no stacking. Default-OFF where it touches core callers.

ID	Deliverable	Depends	Gated?	Status	PR
M0	This design doc + project dir + `runlogs/` + ADR-014	?	owner sign-off	?	**? APPROVED 2026-06-21**
M1	`deployment.profile` + `compute_target` config + preset bundles + precedence/auto-detect; unit tests (no behaviour change, profile='' default)	M0	safe	?	[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)
M2	`input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri` (local=identity); storage-fake tests	M1	safe	?	[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280)
M3	Configurable output/scratch base ? kill hardcoded `output/` (`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden + stub-E2E byte-identical	M1	? default-OFF + Launch Report on flip	?	[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282)
M4	`DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring	M1	safe (cuda default unchanged)	?	[#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283)
M5	**truly-local profile** end-to-end (preset + startup checks); first committed **runlog** (truly-local sample run)	M2,M3,M4	owner real-GPU	?	[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-checks+L4 parity ?; runlog ? owner GPU)
M6	Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) + **containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests	M5	? owner (GCP spend)	?	[#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287) (shim+image+mocked ?; real build/run = M7 ? owner)
M7	**cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet)	M6	? owner (spend)	?	?
M8	Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds, wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation + `/api/.../cost`	M7	? owner (billing)	?	[#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ? owner/follow-up)
M9	**Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`; DPA/consent gate wiring	M7	? owner (privacy/counsel)	?	[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; **+ exposure-safety boot posture** in `startup.py` ? edge-auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ? owner/counsel)
M10	`byo_worker` / zero-infra-BYO ? **design-only, DEFERRED**	M8,M9	? explicit owner approval	?	?
M11	**Data-flywheel harness (D12):** on a consented adult job, **capture** the upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent gate (M9).	M7,M9	? owner (consent/privacy)	?	**skeleton landed** (see changelog): `backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook,

consent faked-deny; live activation ? owner/counsel (consent schema) + owned-model/golden data |
| **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the source from + writes the stitched reel **back to the user's Google Drive next to the source** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | **backend landed** (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` + `gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app ? owner** |

Testing strategy is a first-class deliverable ? [`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ? [`runlogs/`](runlogs/).

8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:
1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to proxy).
5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data; counsel-gated to M9).
6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until the owned model clears the floor; capture?store?eval?goldenize meanwhile).

? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:
2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ? distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only** (speed, no output-byte change); NVENC-encode deferred behind a config knob.
9. ? **Input cap ? D16:** **hard-reject `<2GB`** for v1, as a config cap (raise to 4GB later in one line).
10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.

9. Definition of done
- [ ] M1?M4 land (config + input + output-base + device layer), each green on golden cross-OS + experiment no-regression.
- [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical windows + segment ranges.
- [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog / DEPLOY_REPORT** with a real sample run.
- [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth (M9) gates multi-tenant.
- [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented adult jobs; the owned model is on a measured path to the D11 floor.
- [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.

10. References
ADR lineage: ADR-005/008/009/010/011/012/013 ? **ADR-014** (`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`, `HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`, `LOCAL_APPLIANCE_ARCHITECTURE.md` (khelsutra). **Archived predecessor:** `docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`, `stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`, `backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`, `backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-mappers ? synthesis; 58 findings).

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158 ### Changelog
159 - 2026-06-21 ? created from the design workflow output; M0 in progress.
160 - 2026-06-21 ? **M0 APPROVED**; **M1 landed** ([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)): `deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion (D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach). Suite 1068 green, cov 84.11%.
161 - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280)): `input_backend`/`input_root` + URI-aware main API/CLI input via `resolve_input_uri`/`acquire_local_input` (local = identity passthrough, byte-for-byte unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report provenance. Suite 1105 green, cov 84.84%.
162 - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282)): configurable output/scratch base via `backend/utils/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under `output.output_dir` (default `"output"`, byte-identical; default-OFF ? no Launch Report). Fixes the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
163 - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"` (cuda-if-available-else-cpu); explicit `"cuda"` stays strict (no silent slow-CPU downgrade), default unchanged. `"auto"` resolves before argv (never reaches `wasb_infer.py`); CUDA probe cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126 green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
164 - 2026-06-21 ? **M5+ batch authorized + design picks locked** (owner): the remaining §8 picks resolved ? **D16** (M6 = Cloud Run **Jobs** + bucket-poll; **libx264 encode + NVDEC decode-only**; **hard-reject `<2GB`** input cap; dir name confirmed) and **D17** (real cloud verification budget = **$100/?10k cap**, delete-after). Owner gave a blanket go to write+merge the **default-OFF code** for every remaining gated milestone (M5,M6,M8,M9-auth,M9-consent-cols,M11-plumbing,M12), one isolated PR each, reserving the gate for the real run/spend/calibration/counsel. Grounded by gate-map workflow `wf_4849400f`.
165 - 2026-06-21 ? **REAL M7 cloud GPU run ?** (owner-authorized, within the D17 cap): provisioned an **L4** (`g2-standard-4`, asia-southeast1-a; Mumbai stocked-out) ? native WASB on the L4 ? **22 rallies** on the smoke clip ? `gs://?/outputs/05e0e577?/{intervals.csv,report.json}`; VM DELETED ($0, ~$0.21). ? the no-Gemini cloud run needs `--set indexing.skip_ai_handoff=true` (the fully-local path emits WASB candidate windows directly ? the long-flagged "0 rallies" cold-start fix). The reel **stitch** validated locally (`VideoCompiler` ? 22-segment highlights, 85 MB). DEPLOY_REPORT runlog = the cloud-GPU dry-run guide (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`).
166 - 2026-06-21 ? **M9-auth code landed** ([#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)): `backend/api/auth.py` ? Cloudflare Access **Cf-Access JWT verify** (RS256: sig+exp+iss+aud against the team JWKS; lazy `PyJWT[crypto]`; injectable JWKS ? fully-offline tests) + `resolve_tenant` ? tags `create_job(owner_id=?)`; `SecurityConfig.{edge_auth_enabled=False,cf_team_domain,cf_access_aud}` + CORS via `RALLY_CORS_ALLOW_ORIGINS`. **Default-OFF.** Security review (`wf_9b4c71d0`) found NO bypass; locked alg=none + HS256-confusion rejection. **? M9 ? ? the consent cols + ToS/DPA are owner/counsel; the consent migration is now v7 (M8 took v6).**
167 - 2026-06-21 ? **M8 code landed** ([#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288)): per-job **cost ledger** ? `backend/billing.py` (pure `usage × pricebook` math, USD internal / INR display) + a **v6 migration** (NULLABLE cost columns on `jobs` + a versioned `pricebook` table seeded `placeholder-v0` at $0; additive-NULL, idempotent) + `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET /api/jobs/{id}/cost` + `/api/videos/{id}/cost` + `BillingConfig` (**default-OFF**) + a gated `_maybe_record_job_cost` hook. Adversarial review (`wf_498b24af`) folded a **major silent-GPU-under-bill** (an unpriced `gpu_type` is now surfaced, not invisible $0) + the **honest no-op** caveat (usage emission is a follow-up ? cost 0 until wired) + breakdown reconciliation. Suite green, ruff+mypy clean. **? M8 ? ? owner inserts real GCP prices (a new pricebook version) + the worker usage-emission is a follow-up; placeholder ships $0 = no user-facing change. v6 is the shared bump (M9-consent cols extend it).**
168 - 2026-06-21 ? **M6 code landed** ([#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run **Job** (injectable client) then **bucket-polls** a done-marker via `execution_policy.poll_until` + the storage layer (D16); `get_compute_backend` returns a backend ONLY for `compute_target=cloud_run` (else `None` ? the worker's in-process path, **default-

OFF**). The dispatched container runs **inline** (forced `local_gpu`+native, never re-dispatch); the worker writes a `--done-uri` completion marker on success. `deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env; weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf_87b22fa4`) folded a **blocker** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. **? M6 ? ? the real build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).**
169   - 2026-06-21 ? **M5 code landed** ([#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286)): `backend/config/startup.py` ? per-profile fail/warn-fast startup checks (cloud-serving + non-cloud storage = the one fatal coherence error; GPU/creds mismatches = warnings, the runner healthcheck stays the authority), wired into the server (`_enforce_startup_or_exit` ? exit 1) + the worker (`cmd_infer`/`cmd_train` ? rc 2), default-OFF (probe gated behind an active profile ? the worker stays torch-free at `profile=""`). **L4 profile-parity test** (`tests/test_profile_parity.py`): same stub trajectory ? byte-identical windows + segment ranges across truly-local / cpu-mock / no-profile + worker-vs-in-process. Adversarial 4-lens review folded (`wf_34671ac1`). Suite green, ruff+mypy clean. **? M5 ? ? the real-GPU truly-local runlog (L5) is owner-side** (template pre-staged in `runlogs/`).
170   - 2026-06-21 ? **M9 hardening: exposure-safety boot posture** (alpha-shareable "tunnel-to-a-box" enabler): extended `backend/config/startup.py` with a **server-only exposure posture** (new `include_exposure=True` from `_enforce_startup_or_exit`; the worker stays unaffected ? it reads `--video-uri`, not untrusted tenant paths). Gated on `security.edge_auth_enabled` (**default-OFF ? strict no-op**, byte-identical to today): when the owner flips edge auth ON to expose the engine behind the CF-Access gate, three checks fire at **boot** ? `EXPOSED_NO_VIDEO_ROOT` (**fatal**: `allowed_video_root` jail unset ? a tenant could read an arbitrary local file), `EXPOSED_EDGE_AUTH_INCOMPLETE` (**fatal**: CF team-domain/AUD missing ? lifts the current first-request 401 to boot), `EXPOSED_AUTH_EXTRA_MISSING` (**warn**: `PyJWT[crypto]` not importable ? would 500 per request). So a half-configured exposure fails fast instead of leaking/500-ing. +15 unit tests (default-OFF no-op, worker-ignores, both-fatals-compose-with-profile-checks, server-exit, WARN-only-no-exit). Adversarial 3-lens review (`wf_35b735a9`) confirmed default-OFF byte-identical + no bypass; folded a **whitespace-drift** (aligned `auth.resolve_tenant` to `.strip()` team/AUD so "configured" means the same at boot + at request time) + named the config fields in the edge-auth-incomplete message. Pairs with the appliance **T3** network posture (raw `:8000` refused ? owner-run). Suite green, ruff+mypy clean. **? backstops M9 ? the actual exposure flip + CF team/AUD/ingress + key rotation stay owner-side. (Deferred to the PR-2 runbook: document that `upload_dir` must sit under `allowed_video_root`, and that you MUST set `edge_auth_enabled=True` when exposing ? these checks only fire then.)**
171   - 2026-06-22 ? **M11 skeleton landed** (data-flywheel harness, D12): `backend/flywheel.py` (`capture_job` ? per-video `VideoWorkspace` via `backend.storage.put`; `shadow_eval` ? `experiment.compare`/`compare_unlabeled`; `propose_promotion` ? `promotion.promote_if_served_safe` with Gemini=control/owned=proposal; `run_for_job` orchestrator) + `FlywheelConfig` (**default-OFF**) + a gated `_maybe_run_flywheel` job-completion hook mirroring M8's cost hook. **TWO independent guards keep it INERT in production:** `flywheel.enabled=False` (default) AND `consent_attested` is a **FAKED hard-deny** (returns False for every job) ? so even enabled it captures **nothing** until the real consent schema lands. This is the PLUMBING only; it reuses the real workspace/eval/promotion/storage code so it works the day consent + the owned model + golden labels are wired. **NO DB migration** (the skeleton persists no consent). +13 tests (default-OFF no-op, enabled-but-consent-denied no-op, consented-capture writes the manifest, delegation of shadow-eval/promote, the hook is best-effort). Suite green, ruff+mypy clean. **? M11 ? ? LIVE activation is owner/counsel (the consent ToS/DPA + schema ? the consent columns) + owner/GPU (the owned model + golden corpus for shadow-eval/promotion). ? the future M9-consent migration is now `v9` (v7 = `locale`, v8 = the compute-picker `compute` column ? the schema ladder drifts as siblings land; check `PRAGMA user_version` before authoring).**
172   - 2026-06-22 ? **M12 backend landed** (`gdrive-api://` OAuth/API StorageBackend, D14 Northstar): `backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` ? the headless **Drive-API** round-trip (read source + write reel back), distinct from the mount-only `gdrive://` (left untouched). Path-keyed `gdrive-api://<path>` resolves each component ? a Drive file id via `files().list`; implements `fetch_to_local` (`get_media().execute()` bytes), `put` (folder-creating `files().create` + a `_media_body` seam), `exists`/`stat`/`list`/`delete`. Wired the scheme into `ref.py` (`_KNOWN` + the path-keyed branch), `__init__._backend_class`, and the `StorageConfig.backend`/`input_backend` Literals + a `gdrive_api` settings dict (read via a **hyphen?underscore** lookup in `get_storage`). **All Google libs lazy-imported** (`import backend.storage` never pulls them); a `service` is **injectable** ? fully-offline tests via an in-memory fake Drive (round-trip + scheme + registry + the underscore-settings lookup);

`_build_service` (ADC over the Drive scope) raises a clear `storage-gdrive-api`-extra error when libs/creds are absent. **NO DB migration.** Adversarial 2-lens review (`wf_ae8c9b0f`). Suite green (1466), ruff+mypy clean. **? M12 ? ? the LIVE per-user interactive OAuth app (consumer "connect your Drive": client id/secret + refresh tokens) is owner-gated; the backend ships the service-account/ADC build + the fake. Reads no secrets from config (creds via ADC only).**
173

4. user (2026-07-08T05:45:10.748Z)

```
total 116
drwxr-xr-x 1 avidu 197609     0 Jul  8 10:49 .
drwxr-xr-x 1 avidu 197609     0 Jun 25 16:24 ..
-rw-r--r-- 1 avidu 197609  4860 Jul  4 06:58 DEPLOY_REPORT_cloud-serving_2026-07-04.md
-rw-r--r-- 1 avidu 197609  9398 Jul  4 13:58 DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md
-rw-r--r-- 1 avidu 197609  8687 Jul  4 17:11 DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md
-rw-r--r-- 1 avidu 197609  8207 Jul  7 09:03 DEPLOY_REPORT_cloud-serving_alpha-
all1v1_2026-07-06.md
-rw-r--r-- 1 avidu 197609  8094 Jul  5 20:46 DEPLOY_REPORT_cloud-serving_cp-
rebuild_2026-07-05.md
-rw-r--r-- 1 avidu 197609  8474 Jul  7 19:17 DEPLOY_REPORT_cloud-serving_gpu-resident-
nvdec_2026-07-07.md
-rw-r--r-- 1 avidu 197609 21259 Jul  8 10:49 DEPLOY_REPORT_cloud-serving_validation-
latency_2026-07-07.md
-rw-r--r-- 1 avidu 197609  3609 Jul  7 23:11 README.md
-rw-r--r-- 1 avidu 197609 11336 Jul  7 23:11 RUNLOG_cloud-serving_exec-status-
fastfail_2026-07-07.md
-rw-r--r-- 1 avidu 197609  2698 Jun 25 16:24 RUNLOG_truly-local_TEMPLATE.md
```

5. user (2026-07-08T05:45:19.169Z)

```
1       # runlogs/ ? sample runs & deploy reports for posterity
2
3       This directory is the **durable record of real runs** of the compute-decoupled serving
profiles ?
4       the same convention proven by the archived
`DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md`. The
5       goal: **the next deploy is one-shot**, and anyone can trace exactly what was run, what
it cost, and
6       what came out.
7
8       ## When to add a file here
9       - Any **real-hardware run** (owner-side L5 in
[`../TESTING_STRATEGY.md`](../TESTING_STRATEGY.md)) of a profile.
10      - Every **default flip** that touches the core ships its Launch Report record here too
(per [[launch-report-convention]]).
11
12      ## Naming
13      `DEPLOY_REPORT_<profile>_<YYYY-MM-DD>.md` ? e.g. `DEPLOY_REPORT_truly-
local_2026-07-01.md`,
14      `DEPLOY_REPORT_cloud-serving_2026-07-01.md`.
15
16      ## Required contents (mirror the DECOUPLED deploy report)
17      1. **TL;DR** ? what ran, profile, ?/?/? per stage (detect / window / stitch / serve).
18      2. **Resources** ? exact machine/VM/Cloud-Run config, region, image; for cloud, the
bucket + SA + budget.
19      3. **Exact commands** ? copy-pasteable (`worker infer ?`, the dispatch call, the
config/profile used).
20      4. **Run results** ? rally counts, **profile-parity numbers** (windows/segments vs the
other profile),
21         and the metering trio **`gpu_active_seconds` / `wall_seconds` / `bytes_out`** (even
before the M8 ledger lands).
22      5. **Reel smoke-test** ? `ffmpeg -f null -` full decode (exit 0) + a 5-frame contact
sheet visually
23         confirming real rallies (court/net/players mid-rally).
24      6. **Gotchas** ? conda-ToS, WASB-C3, GPU stockouts, font fallback, cold-start, NVDEC,
```

```
etc.
7. **Cost** ? measured/estimated; reconcile probe metrics vs the platform's per-invocation billing.
8. **Teardown / final state** ? for cloud, the `$0` audit (`gcloud compute instances list` + `disks list` empty); **DELETE, never just stop** ([[gcp-vm-always-delete]]).

## DoD coupling
`COMPUTE_DECOUPLED_SERVING` is not DONE until **`truly-local` (M5)** and **`cloud-serving` (M7)** each
have **?1 committed report here** (README §9).

## Index (append as runs land)
| Date | Profile | Result | File |
|---|---|---|---|
| 2026-07-06 | cloud-serving | alpha-tester "All 1 v 1" timed out ? downsample fix proven on L4 (720p30: 956 s, 60 rallies) | [DEPLOY_REPORT_cloud-serving_alpha-all1v1_2026-07-06.md](DEPLOY_REPORT_cloud-serving_alpha-all1v1_2026-07-06.md) |
| 2026-07-07 | cloud-serving | GPU-resident NVDEC decode (#506/#507) shipped: image bump + cutover, cpu/nvdec smokes, GX010128 trajectory reproduced VM (26479?26481 vis), enabled + canaried | [DEPLOY_REPORT_cloud-serving_gpu-resident-nvdec_2026-07-07.md](DEPLOY_REPORT_cloud-serving_gpu-resident-nvdec_2026-07-07.md) |
| 2026-07-07 | cloud-serving | Crash-before-marker fast-fail (#515) LIVE smoke: real `get_execution` classifier validated vs history (Succeeded+Failed); crash?**rc 5 in 47 s** (not ~65 min); healthy dispatch **rc 0 / 19 rallies**, probe returned None on all 9 running-polls | [RUNLOG_cloud-serving_exec-status-fastfail_2026-07-07.md](RUNLOG_cloud-serving_exec-status-fastfail_2026-07-07.md) |
| 2026-07-07 | cloud-serving | validation-latency closure (#512 shared-decode + #513 worker skip-reval): worker A/B on live image proves Video 1 FAIL?PASS (13 rallies) + re-val 169 s?70 s (2.43×) on L4; off-box #512 1.95?3.23× across codecs, zero decision drift; CP baseline 6m52s (live rev-00018 number owner-gated) | [DEPLOY_REPORT_cloud-serving_validation-latency_2026-07-07.md](DEPLOY_REPORT_cloud-serving_validation-latency_2026-07-07.md) |

```

6. user (2026-07-08T05:45:19.573Z)

```
<!--
PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.

A "tracked project doc" is how this repo carries substantial design/project work: it can be
ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables, then
archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
docs/GOLDEN_REGRESSION_FIXTURES.md.

Rules of thumb:
- §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable PR.
- Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs sign-off),
  or [design] (proposed). The doc reflects real shipped state, never aspirational.
- It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
- Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model for
  research- or security/privacy-heavy work). Delete sections that do not apply.
- Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state work).
-->

# <Project / Plan Title>

> **Status:** `DRAFT` ? `<owner-gated? | in progress>` · **Owner:** `<name>` · **Created:** `<YYYY-MM-DD>` · **Last updated:** `<YYYY-MM-DD>`
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
```

```
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
> **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
> **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not legal/financial advice where applicable.

---

## 0. TL;DR
<2?5 sentences: the goal, the approach, and the single most important caveat. Call out any finding that is already actionable on its own.>

## 1. Problem & goal
<Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a reader must read to get current.>

## 2. Decisions locked
<Table ? capture answered questions so they are not relitigated.>

| # | Decision | Source / date | Implication |
|---|----------|---------------|-------------|
| D1 | ? | ? | ? |

## 3. Foundation ? research / prior art  *(OPTIONAL)*
<Include when the plan rests on external facts: cited research, legal/compliance, benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this creates.>

## 4. Design / architecture
<The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing modules/code this builds on so it is clear what is new vs reused.>

## 5. Threat model / risk table  *(OPTIONAL ? security / privacy / compliance work)*
<Who/what is protected vs NOT; residual risks stated plainly.>

## 6. Honest limits ? what this does NOT do
<The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>

## 7. Deliverables & progress tracker   ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

| ID | Deliverable | Depends on | Gated? | Status | PR |
|----|-------------|-----------|--------|--------|----|
| P0 | ? | ? | No | ? | ? |

## 8. Open questions ? owner / external  *(split blocking-gates from nice-to-knows; name who decides)*
1. ?

## 9. Definition of done
<Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
- [ ] ?

## 10. References
**Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`. **External:** `<cited sources>`.

---

### Changelog
- `<YYYY-MM-DD>` ? `<what changed>`.
```

7. user (2026-07-08T05:45:26.555Z)

DEPLOY REPORT ? cloud-serving, validation-latency closure (#512 shared-decode + #513 worker skip-revalidation), 2026-07-07

> Durable **closure record** for the two validation-latency fixes now live on the `cloud-serving` path: **#512** (`perf(validators)`: decode `scene_cut`/`blur`/`relevance` in ONE shared pass instead of ~125 random per-sample seeks across 3 captures) and **#513** (`fix(serving)`: the worker skips its redundant re-validation when the control-plane already validated + dispatched, via `--skip-validation`). Both shipped in the live image `rally-worker@sha256:8045f54e?` (built from `origin/master` @ `f5c9dd9`; #512 `20862d7`, #513 `23c1868` are ancestors). Measures the before/after on 3 clips spread across codecs/resolutions, all against the real serving infra. Project `gen-lang-client-0306026826`, region `asia-southeast1`. Honest-limits (incl. the owner-gated live CUJ) in §6.

1. TL;DR

Stage	Result
Live image carries #512+#513	? control-plane rev `00018-qvz` + worker Job both on `sha256:8045f54e?` (= `f5c9dd9`, ancestors `20862d7`/`23c1868` verified)
Baseline captured (pre-fix)	? from prod logs on rev `00017-qbd` (image `09d2edf7` = `66756ad`, pre-#512/#513) ? no re-run needed
#513 worker skip-revalidation	? real dispatch-style exec: Video 1 validation **SKIPPED (0 s)**, runs to **SUCCESS / 13 rallies** ? the blur-reject that failed the live CUJ is **gone**
#512 shared-decode speedup (real L4)	? Video 1 re-validation **169.3 s ? 69.7 s (2.43×)** on identical worker hardware, decision byte-identical (sharpness 11.9 < 20.0 both)
#512 speedup (off-box A/B, breadth)	? 1080p H.264 **1.95×**, 4K HEVC **3.10×**, 5.3K 10-bit HEVC **3.23×**; every validator decision unchanged OLD?NEW
Live control-plane "after" number	? **live-corroborated 2026-07-08** (§9): rev `00019` validated a fresh 1.83 GiB / 174 s clip in **183 s** (in the 2?3 min band) + reel compiled end-to-end. Video-1's exact 412 s?~170 s A/B stays a projection (the live CUJ used a different, heavier clip).

Bottom line: the two fixes do exactly what they claim on live serving hardware. **#513** turns the soft 5.3K clip from a **hard FAIL** (worker re-validated at its default `min_blur=20`, rejected sharpness 11.9, `rc2`, no reel) into a **PASS with 13 rallies** by trusting the control-plane's gate. **#512** cuts the frame-sampling validators ~**2?3×** (2.43× measured on the L4; 1.95?3.23× off-box across codecs) with **zero decision drift** OLD?NEW. The only number still owner-pending is the live control-plane validation wall on rev 00018 ? its components are measured and it projects to ~2?3 min from the ~6m52s baseline.

2. Resources
- **Control-plane:** `rally-control-plane` Cloud Run **service**, rev **`rally-control-plane-00018-qvz`** (100% traffic), image `asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:8045f54ea9fcbf?` (built from `f5c9dd9`), `DEPLOYMENT_PROFILE=cloud-serving`, CPU-only **2 vCPU**, edge-auth ON (`edge_auth_enabled=true`, `cf_team_domain=khelsutra.cloudflareaccess.com`).
- **Worker:** `rally-worker` Cloud Run **Job**, same image `sha256:8045f54e?`, 8 vCPU / 32 GiB / 1× NVIDIA **L4**, `WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar`, `WASB_DECODE_BACKEND=nvdec_resident`, GCSFuse mount of `gs://khelsutra-rally-serving`.
- **Baseline (pre-fix) surface:** control-plane rev **`00017-qbd`** / worker image `sha256:09d2edf7?` (= `66756ad` = #507+#508+#509, both **pre-#512/#513**) ? the source of the prod baseline logs below (2026-07-07, before the 16:15 rev-00018 cutover).
- **Corpus:** Video 1 `gs://khelsutra-rally-serving/videos/uploads/702636eed76145e88aa59c3000c7e415/Video 1 - Badminton.MP4` (1.41 GiB, 5312×2988 yuv420p10le HEVC, 3019 frames, MD5 `8fe6b719603dd7a7b2b6ef543139a4a3`); `gs://khelsutra-rally-corpus/videos/golden_1080p/Boxhill_Doubles.mp4` (1080p H.264); `gs://khelsutra-rally-corpus/videos/golden/GX010128.mp4` (2.18 GiB, 4K HEVC, 45298 frames).

3. Exact commands
```bash
# --- Provenance: prove the live image carries both fixes ---
gcloud run services describe rally-control-plane --region asia-southeast1 \
  --format="value(status.latestReadyRevisionName, spec.template.spec.containers[0].image)"   # 00018-qvz  ?@sha256:8045f54e?
```

```
git merge-base --is-ancestor 20862d7 f5c9dd9 && echo "#512 in live image"   # yes
git merge-base --is-ancestor 23c1868 f5c9dd9 && echo "#513 in live image"   # yes

# --- #513 proof: worker A/B on the LIVE image (Video 1 already staged in the serving bucket) ---
V1="gs://khelsutra-rally-serving/videos/uploads/702636eed76145e88aa59c3000c7e415/Video 1 - Badminton.MP4"
# Run A ? WITH --skip-validation (exactly what orchestration._maybe_dispatch_remote emits on dispatch):
gcloud run jobs execute rally-worker --region asia-southeast1 --async \
  --args="^@@^infer@@--video-uri=$V1@@--out-uri=gs://khelsutra-rally-serving/_valclose/v1-skipval@@--segmenter=wasb_hybrid@@--skip-validation@@--set=deployment.compute_target=local_gpu@@--set=indexing.detector_impl=native@@--set=indexing.skip_ai_handoff=true@@--set=indexing.wasb.batch_size=64@@--set=indexing.wasb.prefetch_batches=4@@--set=indexing.wasb.timeout_sec=0"
# Run B ? WITHOUT --skip-validation (worker re-validates with the NEW #512 validators ? still blur-rejects):
gcloud run jobs execute rally-worker --region asia-southeast1 --async \
  --args="^@@^infer@@--video-uri=$V1@@--out-uri=gs://khelsutra-rally-serving/_valclose/v1-reval@@--segmenter=wasb_hybrid@@--set=deployment.compute_target=local_gpu@@--set=indexing.detector_impl=native@@--set=indexing.skip_ai_handoff=true"

# --- #512 timing: bracket "pulled ? -> in.mp4"  ?  "validation FAILED/SKIPPED"  ? "Phase 2 (WASB) completed" ---
gcloud logging read 'resource.type="cloud_run_job" AND labels."run.googleapis.com/execution_name"="rally-worker-<exec>"' \
  --freshness 1d --format="value(timestamp, textPayload)"

# --- #512 off-box A/B (isolated clones; run_all decodes the SAME sample indices in both trees) ---
git clone --local --no-hardlinks <repo> repo_old && git -C repo_old checkout 23c1868   # pre-#512 (per-sample seeks)
git -C repo_old worktree add --detach repo_new f5c9dd9   # live image src (shared pass)
python valbench.py <repo_old|repo_new> <video> <label> 2   # times registry.run_all(video, load_config())
```

4. Run results

4a. Video 1 (5312×2988 10-bit HEVC, 3019 frames) ? the clip that motivated #512/#513

Metric	BEFORE ? pre-fix (rev `00017`, img `09d2edf7`)	AFTER ? #512+#513 live (rev `00018`, img `8045f54e`)
Control-plane validation	**411?412 s** (6m52s) ? passed (baked `min_blur=8.0`)	**projected ~170 s (2?3 min)**; **live-corroborated 2026-07-08 ? 183 s** on rev `00019` (§9 ? a fresh 1.83 GiB clip, not a Video-1 re-run)
Worker (re-)validation	**169.3 s** ? ? **BLUR-REJECT** (sharpness 11.9 < default 20.0)	**0 s ? SKIPPED** (`--skip-validation`, #513)
? counterfactual: worker re-val if NOT skipped	*(169.3 s, OLD validators)*	**69.7 s** (NEW #512 validators, **2.43×**, same 11.9<20 decision)
End-to-end	~10m31s wall ? **FAILED** (0 rallies, no reel)	worker ? **SUCCESS** in ~2m38s exec ? **13 rallies**; **reel live-confirmed 2026-07-08** (§9 ? fresh-clip CUJ compiled an 11-clip reel end-to-end)
Rallies produced	**0** (rejected before detection)	**13** (`report.json` `status=success`, 13 segments)
Pass / fail	? **FAIL**	? **PASS**

Prod baseline evidence (rev 00017, 2026-07-07, unmodified): control-plane `13:49:18 Running validations ? 13:56:10 Video passed checks` = 411.4 s (and `14:03:56 ? 14:10:48` = 411.9 s). Worker exec `rally-worker-2lbz2` (dispatched by the 14:03 run, OLD image): `pulled 14:11:35 ? validation FAILED: blur_check ? sharpness 11.9 below 20.0 @14:14:24 ? exit(2)` ? a **successful GPU dispatch that failed after the fact**, the exact CUJ #513 fixes.

After evidence (live image `8045f54e`, 2026-07-07):
- `rally-worker-k29rk` (**Run A, --skip-validation**): `pulled 17:09:24 ? [worker] validation SKIPPED (--skip-validation) 17:09:27 ? native WASB (torch 2.6.0+cu124, cuda True) ? Phase 2 (WASB) completed in 100.9 s, 13 candidate windows (1510/3019 visible) ? emitted 13 rallies ? exit(0)`. Exec wall 157.7 s; `gpu_active`?100.9 s (WASB); `bytes_out`=`report.json`+`intervals.csv`. Output `gs://?/_valclose/v1-skipval/outputs/8fe6b719?/`, `status=success`, 13 segments.
- `rally-worker-t8llx` (**Run B, re-validation on the NEW image**): `pulled 17:09:29 ? validation FAILED: blur_check ? sharpness 11.9 below 20.0 @17:10:39 ? exit(2)`. Re-val wall **69.7 s** vs the OLD image's 169.3 s = **2.43× on identical L4 hardware**, and the **decision is byte-identical** (11.9 < 20.0) ? #512 is pure speed, zero behavior change. (This run also shows *why* #513 is needed: absent the skip, the worker's default `min_blur=20` still rejects the clip the control-plane passed at 8.0.)

4b. Off-box #512 A/B ? codec/resolution breadth (16-core local box, cv2 4.13.0, warm, `run_all`)

Isolated clones at `23c1868` (OLD, per-sample seeks) vs `f5c9dd9` (NEW, shared pass); identical sample indices ? a fair A/B. **Every validator decision is identical OLD?NEW** on every clip (no PASS/FAIL drift from the fix).

Clip	codec / res	OLD (warm)	NEW (warm)	Speedup	decisions OLD?NEW
Boxhill_Doubles	1080p H.264	10.91 s	5.60 s	**1.95×**	6/6 identical (all PASS)
GX010128	4K HEVC (45298 fr)	35.76 s	11.54 s	**3.10×**	6/6 identical (all PASS)
Video 1	5.3K 10-bit HEVC	173.20 s	53.62 s	**3.23×**	6/6 identical (all PASS?)

? On this Windows/cv2 box the 10-bit-HEVC decode yields a Laplacian variance **above** 20 so blur PASSES off-box, whereas the Linux serving stack measures **11.9** and rejects. That platform decode-stack difference is **orthogonal to #512** (which changes only *which frames are sampled*, identically in both trees); the authoritative blur verdict and the real speedup come from the L4/prod (§4a). Off-box contributes only the OLD?NEW **ratio**, which is decode-stack-independent.

The win grows with per-frame decode cost (random-seek re-decode is what #512 removes), and it **triangulates** with the real-L4 Video 1 A/B (2.43×): the off-box ratios bound the control-plane projection.

4c. Control-plane projection (2-vCPU)
Baseline **~412 s**. Applying the L4-measured **2.43×** ? **~170 s (~2m50s)**; the off-box ratios (1.95?3.23×) bracket **~2?3 min**. Matches the pre-fix design estimate (~6.5 min ? ~2 min). Exact live number is owner-gated (§6).

5. Rollout state
- **Already deployed ? this is a measurement/closure record, not a deploy.** #512+#513 went live with the `8045f54e` image cutover (control-plane rev `00018-qvz`, worker Job same digest) on 2026-07-07. No infra change was made for this report.
- **Rollback (pre-existing levers):** control-plane `gcloud run services update-traffic rally-control-plane --region asia-southeast1 --to-revisions rally-control-plane-00017-qbd=100`; worker `gcloud run jobs update rally-worker --region asia-southeast1 --image ?/rally-worker@sha256:09d2edf7?`. Both revert to the pre-#512/#513 image.
- **Scratch outputs** written under `gs://khelsutra-rally-serving/_valclose/` (isolated prefix; safe to delete).

6. Honest limits / not-verified
- **Live end-to-end CUJ is owner-gated.** ? **CLOSED 2026-07-08 ? see §9** (owner drove one live SPA upload on rev `00019`; both ? cells now measured). Original limit, retained for context: `POST /api/process` on rev 00018 requires a Cloudflare-Access JWT (`edge_auth_enabled=true`; a direct call returns **401** `missing Cf-Access-Jwt-Assertion header`). So the two ? cells ? the **live control-plane validation wall on rev 00018** and the **stitched reel** ? need the owner to drive one upload (or hand over a CF-Access JWT / service token). Everything else here is measured on the real serving infra. **Recipe to close them:**

upload "Video 1 - Badminton.MP4" via the SPA (compute=cloud), then `gcloud logging read '?service_name="rally-control-plane"?("Running validations" OR "passed checks")'` for the wall, and confirm the dispatched worker exec logs `validation SKIPPED` + emits rallies + the reel compiles. Expect CP validation ~2?3 min and a PASS.
94 - **#513's worker win is shown via a dispatch-faithful direct exec**, not a control-plane-initiated dispatch (same auth gate). The `--skip-validation` flag, its emission in `orchestration._maybe_dispatch_remote`, and its handling in `worker.cmd_infer` are unit-tested on master; Run A exercises the exact worker code path with the exact flag. ? **UPDATE 2026-07-08 (§9): now also proven on a genuine control-plane-initiated dispatch** (`rally-worker-64bhj`): `_maybe_dispatch_remote` emitted `--skip-validation` on a live SPA run and the worker honored it (`validation SKIPPED`, 0 re-validation).
95 - **Rallies, not a reel.** Run A used `skip_ai_handoff=true` (secret-free) so the 13 "rallies" are local candidate windows (no Gemini refine, no stitched mp4) ? this validates decode+detect+windowing end-to-end and that the clip **passes**; the reel stitch is the owner-CUJ step above. The 13 segments are its inputs.
96 - **Off-box absolute times are from a Windows 16-core box (cv2/FFmpeg), not the 2-vCPU Linux control-plane** ? so the **ratios** transfer (the change is algorithmic: seeks?shared forward-hybrid pass), the absolutes don't, and (per ?) the 10-bit blur *verdict* can differ; the control-plane absolute is anchored to the prod 412 s baseline, not the local numbers. Video 1 off-box is a cross-check of the authoritative real-L4 2.43×.
97 - **#515** (crash-before-marker fast-fail) is NOT in this image (`ef41473` post-dates `f5c9dd9`); it's unrelated to validation latency and out of scope for this closure.
98
99 ## 7. Cost
100 - 2 L4 worker execs (Run A 157.7 s + Run B 109.6 s ? 268 s of L4 time) ? **~$0.5?0.7**. Off-box A/B: **$0** (local). No VM, no image build. GCS read for the 3 downloaded clips (~5.1 GiB) is intra-project.
101
102 ## 8. Closure verdict
103 ? **CLOSED ? #512 and #513 are proven on live cloud-serving hardware.** #513 converts the motivating soft-focus 5.3K clip from a post-dispatch FAIL into a 13-rally PASS by eliminating the worker's redundant re-validation (169 s + wrongful blur-reject ? 0 s); #512 speeds the frame-sampling validators **2.43× on the real L4** and **1.95?3.23× off-box across codecs**, with **zero decision drift** OLD?NEW on every clip. The last owner-gated residual is now **closed live 2026-07-08 (§9)**: an owner-driven SPA CUJ on rev `00019` validated a fresh 1.83 GiB / 174 s clip in **183 s** (in the projected 2?3 min band) and compiled an 11-clip reel end-to-end ? with the worker `validation SKIPPED` on a genuine control-plane dispatch. (The Video-1-specific 412 s?~170 s A/B stays a projection by design ? the live CUJ used a different, heavier clip.) Validation-latency work is done.
104
105 ## 9. Update (2026-07-08) ? live SPA CUJ closes the two owner-gated cells (on rev `00019`)
106
107 The two ? cells in §4a (the **live control-plane validation wall** and the **end-to-end reel**) were owner-gated on the Cloudflare-Access `POST /api/process`. On **2026-07-08** the owner drove one real SPA upload (compute=**Cloud**) end-to-end, so both are now measured on live serving infra. Mirrors the gpu-resident report's "§5 Update" append ? the body above is unchanged; this section records what the live run added.
108
109 **Two honest deltas from the recipe** (state moved after this report's 2026-07-07 image cutover):
110 1. **Live rev rolled `00018` ? `00019`.** Traffic is 100% on **`rally-control-plane-00019-96f`**, image `rally-worker@sha256:c6393d27b4ae?` (tag `:latest`), deployed 2026-07-07T17:06Z ? ~51 min after the `00018`/`8045f54e` cutover §2 documents; the worker Job is on the same `:latest`=`c6393d27` digest (both surfaces match). It's an undocumented forward-bump, but #512 (`20862d7`) and #513 (`23c1868`) are ancestors of `origin/master`, `f5c9dd9` (the `00018` src) is an ancestor of master, and the run **empirically** shows both fixes live (one-pass validation wall + worker `validation SKIPPED`). Measured on what is actually serving.
111 2. **A fresh clip, not Video 1.** The upload was **"First CUJ.MP4"** ? **1.83 GiB** (1,960,043,219 B), source duration **174.17 s** (?84 Mbps, 4K-class ? *heavier* than Video 1's 1.41 GiB), video_id `2f31b45e?`. So the live wall **corroborates** the Video-1 projection on a comparably-heavy clip; it is **not** a re-measurement of Video 1's 412 s?~170 s A/B (a different clip has a different absolute wall).
112
113 **Full phase breakdown (UTC 2026-07-08, rev `00019`):**

#	Phase	Window	Wall	Machine	Notes
1	Upload (client?server, 80-part multipart)	04:25:56 ? 04:28:51	**~2m55s**	client uplink	1.83 GiB @ ~84 Mbps ? owner bandwidth, not infra
2	Accept ? validation start (stage)	04:28:52 ? 04:29:22	~30 s	control-plane	stage upload to serving bucket
3	**Validation (#512 shared-decode)** ?	04:29:22 ? 04:32:25	**183 s (3m03s)**	control-plane 2 vCPU	one-pass scene_cut/blur/relevance; PASSED
4	Dispatch ? worker ready (cold start)	04:32:25 ? 04:33:20	~55 s	Cloud Run Job	image pull + GCSFuse mount + torch/cuda init
5	**Worker skip-validation (#513)** ?	04:33:20 ? 04:33:24	~4 s	L4	`validation SKIPPED`, 0 re-validation
6	Detection / segment (WASB)	04:33:24 ? 04:36:20	**161.6 s**	L4 (cuda)	15 candidate windows, 2561 visible pts
7	Ingest + finalize	04:36:20 ? 04:36:24	~4 s	control-plane	intervals?DB, `status=success`
?	(user reviews rallies, clicks Compile)	04:36:24 ? 04:37:26	~62 s idle	?	human-in-the-loop
8	**Compile / stitch** ? slow	04:37:26 ? 04:50:21	**~12m55s**	control-plane 2 vCPU	11 clips, ~67 s/clip 4K re-encode+watermark; concat ~2 s; mirror ~11 s

Server pipeline (process?success, excl. upload+idle): **7m32s**. Full CUJ (upload?reel): **~24m25s** wall.

Cell 1 ? control-plane validation AFTER (live): **183 s (3m03s)** on rev `00019`, PASSED ? the #512 shared-decode validators on the real 2-vCPU serving path, in the projected **2?3 min** band on a clip *heavier* than Video 1 (pre-#512 per-sample seeks were the ~6m52s-class baseline). Evidence: `Running validations 04:29:22.789 ? Video passed checks 04:32:25.780`.

Cell 2 ? end-to-end reel (live): `/api/compile` stitched **11 of the 15** detected rallies (0.5 s pad, per-clip branding watermark) from the 174.17 s source ? **`gs://khelsutra-rally-serving/highlights/2f31b45e?/first-cuj-ks-highlight-reel.mp4`** (**581.84 MiB**, saved 04:50:10 *"Dynamic highlight reel saved (581.8 MB)"*, mirrored 04:50:21, served via `/api/highlights` 307). **Reel ? produced end-to-end on the live control-plane** ? source fetch ? per-clip trim+watermark ? concat ? mirror-upload, all exercised.

Bonus ? #513 proof upgraded from the §6 dispatch-faithful *direct exec* to a genuine **control-plane-initiated dispatch** (`rally-worker-64bhj`): `_maybe_dispatch_remote` emitted `--skip-validation` on a live SPA run and the worker honored it (0 re-validation).

? Finding ? stitch is the scaling bottleneck (follow-ups filed). The compile ran **~13 min on the CPU-only 2-vCPU control-plane** (~45?90 s per ~7 s clip: the watermark forces a full 4K re-encode, no NVENC). With validation (183 s) *also* on the control-plane and serialized by `_LOCAL_COMPUTE_LANE`, one run pins the control-plane for ~16 min of heavy CPU ? it won't scale past a few concurrent users. Filed: **#521** (move compile/stitch to the L4 worker + make phase?machine placement config-driven) and **#520** (step-level debug tracing, server+UI, to a low-latency sink ? this breakdown had to be hand-correlated across 3 log streams).

Cost (heuristic): one L4 detect exec (`rally-worker-64bhj`, ~222 s) ? **$0.15?0.55** (rate-dependent, dominant) + control-plane CPU for validation+stitch (~960 s @ 2 vCPU) ? **~$0.05** + reel egress (581 MB, if downloaded to internet) ? **~$0.07** + storage/ops **< $0.02** + Gemini **$0** (skip_ai_handoff). **? $0.3?0.7 total**, ~one L4 exec as projected ? but the stitch's real cost is **concurrency** (~16 min of control-plane heavy CPU), not dollars.

Net: both owner-gated cells are now live-measured on rev `00019` ? validation wall **183 s** (corroborating the 2?3 min projection) and the reel compiles end-to-end. #512 one-pass validation and #513 skip-validation-on-real-dispatch are now both directly observed in prod. The Video-1 412 s?~170 s A/B remains a projection by design (different, heavier clip).

8. user (2026-07-08T05:45:27.116Z)

Documentation Status & Archival Register

> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.
> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).

---

0. How to use this
- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.

1. Lifecycle legend
`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
Before a foundational/tracked doc moves to `docs/archives/`:
1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.

3. The register

3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
Doc	Status	Archive-ready?	Next action
`ALPHA_LAUNCH_READINESS.md`	**DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates. A0?A9 DONE; A1 live (Cloud Run, e2e runlog); **A13 review DELIVERED 2026-07-04** (`ALPHA_LAUNCH_REVIEW_2026-07-04.md`)	No	Owner: record GO/NO-GO (§10) + close/defer A10/A11/A12; fix khelsutra #154 copy blockers; run the 5-min browser-CUJ checklist (review §1.1)
`COMPUTE_DECOUPLED_SERVING/`	**IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`)	No	M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)
`RALLY_QUALITY_RESEARCH.md`	Foundation, **tracked & ACTIVE** (§7)	No	Drives the R-series; complete when §7.1 single quality number is attributed

36 | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
37 | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |
38 | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |
39 | `RALLY_DETECTION_FIXES_PLAN.md` | **DONE** ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + §7.1 number) | Archive with the R-series when the track wraps |
40 | `RALLY_DETECTION_CODE_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |
41 | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |
42 | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |
43 | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |
44 | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |
45 | `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |
46 | `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |
47 | **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`. |
48 | `archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md` **(umbrella)** | **? COMPLETE / ARCHIVED (2026-07-02)** ? the five-repo code audit + folded-in *Remediation tracker* (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved to `archives/past_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |
49 | `DECISION_SNAPSHOT_REGRESSION.md` | **DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |
50 | `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |
51 | `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |
52 | `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |
53 | `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
54 | `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |
55 | `POST_166_...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |
56
57 ### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")
58 | Doc | Status | Note |
59 |---|---|---|
60 | `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |
61 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
62 | `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
63 | `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)** |
64 | `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |
65 | `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |

| `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
| `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

3c. Reports & terminal records (archive candidates once their track wraps)
Doc	Status	Note
`ALPHA_LAUNCH_REVIEW_2026-07-04.md`	**REPORT ? DELIVERED 2026-07-04** (A13) ? owner-facing alpha launch review: GO/NO-GO recommendation (conditional GO for the current 4-email allowlist, gated on khelsutra #154 copy fixes + the owner browser-CUJ checklist; NO-GO for wider invites until A10/A11 + SA least-privilege + retention), DoD scorecard, operator runbook, observability/triage incl. the 2026-07-04 live failure-mode battery, privacy/consent audit, known limits + rollback	Awaiting the owner GO/NO-GO record on `ALPHA_LAUNCH_READINESS.md` §10; archive with the alpha cluster once recorded
`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING` · `HARDENING_LOOP_HANDOFF` · `DEFERRED_HARDENING_PLAN`	COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the umbrella's *Prior audits*)	? Full copies under `docs/archives/*-COMPLETED-2026-06-14.md`; inbound links repointed (finding #4 resolved)

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)
`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

3e. Already archived (`docs/archives/`)
`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` · `past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

3f. ADR log ? `docs/archives/decisions/DECISIONS.md`
ADR-001 ? ADR-014 (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

4. Open due-diligence findings / actions (2026-06-21 review)

> **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected

as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
93   4. **? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.** The three top-level pointer-stubs (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's *Prior audits (absorbed)*.
94   5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
95   6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
96
97   ## 5. Maintenance
98   Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
99

9. assistant (2026-07-08T05:45:32.806Z)

Now let me read the ADRs. Let me find where 012/013/014 live in DECISIONS.md.

10. user (2026-07-08T05:45:34.452Z)

598:## ADR-012: Compute = GPU-native (L4/T4); TPU deferred; Cloud Run + GPU (scale-to-zero) is the supported serving model
661:## ADR-013: Drop DigitalOcean for compute ? GCP is the SOLE compute stack (training + serving)
686:## ADR-014: Compute-decoupled rally serving ? one pure core, profile-selected compute + storage

11. user (2026-07-08T05:45:35.746Z)

714 C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/decisions/DECISIONS.md

12. user (2026-07-08T05:45:41.206Z)

598   ## ADR-012: Compute = GPU-native (L4/T4); TPU deferred; Cloud Run + GPU (scale-to-zero) is the supported serving model
599
600   **Date:** 2026-06-20 · **Status:** RATIFIED (owner, in-session). **Refines ADR-011** (concretizes the "GCP = serving" path).
601
602   ### Decision
603
604   1. **Stay GPU-native (NVIDIA L4/T4) for all compute** ? local box, GPU VM, and the serving path ?
605   for the foreseeable roadmap. **Do NOT invest in TPU now.** Revisit TPU **only** if/when a large
606   *owned* transformer (Qwen3-VL Gen-2+, owner-gated) at *sustained batch* scale becomes the dominant
607   cost, and **only after a benchmark** proves it ? never speculatively.
608   2. **Cloud Run + GPU (L4), scale-to-zero, is the supported SERVING deployment model** for the
609   pay-per-video orchestration economics ("users pay us to orchestrate on our GCS bucket + VMs /
610   Cloud Run"): pay-per-request GPU-seconds = the exact per-video bill, zero idle-GPU cost, co-located

with `gs://khelsutra-rally-corpus`. This is the concrete form of ADR-011's deferred **M3/M4**.
3. **The named optimization lever is GPU-accelerated video decode (NVDEC)**, not more GPU FLOPs ? the measured bottleneck is video *decode*, not the model.

Rationale

- **Workload isn't TPU-shaped.** WASB is a tiny (5.8 MB) PyTorch CNN doing per-frame heatmap inference; the owned model is a VLM. TPUs win on large, dense, *batched* matmul at training scale; streaming per-frame CNN inference with small/dynamic batches is a poor fit.
- **The real bottleneck is video decode, which TPU does nothing for.** Measured on the 2026-06-20 GCP L4 M2 run: cv2 PNG extraction at ~12 fps with the **GPU idle during extraction**. The lever that helps is **NVDEC** (hardware decode on L4/T4) + parallel decode ? i.e. lean *into* the GPU.
- **Portability / single-runtime.** TPU needs XLA (torch_xla / JAX) ? significant fragile NRE to port WASB (torch 1.11) + the pipeline, a second runtime to maintain, and **GCP-only** ? it would break the local ? VM ? Cloud Run parity the DECOUPLED storage/compute seams exist to provide. One portable PyTorch+CUDA image that runs identically everywhere is worth more than peak FLOPs here.
- **Economics.** Cloud Run+GPU scale-to-zero meters per request = clean per-video billing; TPU pricing (v5e pods / queued / committed-use) targets sustained training and its regional availability is *narrower* than GPU (the M2 run already hit a Mumbai GPU stockout ? landed in Singapore; TPU would be worse).

Consequences

- The **deferred M3 (inference serving endpoint + per-video billing) and M4 (scale-to-zero + cost guard)** from ADR-011 become a **new tracked subproject ? "Cloud Run + GPU serving"** ? to be spun up (as a `docs/PROJECT_DOC_TEMPLATE.md`-based doc) **AFTER `docs/DECOUPLED_COMPUTE/` is archived** (owner sequencing). Seed scope below. **? REALIZED in ADR-014 (2026-06-21):** the subproject is `docs/COMPUTE_DECOUPLED_SERVING/`, **renamed** to reflect that **truly-local is a co-equal profile** (not just cloud serving); ADR-014 expands this 7-point seed into the M0?M10 plan.
- **First subproject work item = containerize `python -m backend.worker`** for Cloud Run. This is the known hard part: WASB's torch-1.11 conda env makes the image non-trivial (two runtimes ? the indexer venv + the `wasb` conda env). The decoupled storage/compute seams + the native runner (`detector_impl="native"`, PR #256) make the *code* portable; packaging is the gap.
- Keep the **storage/compute decoupling seams** intact so the SAME container runs local / GPU VM / Cloud Run (the per-video workspace convention, in progress, reinforces this).
- **Edge auth is a hard prerequisite** before any public serving (engine API is CORS `*`, no auth) ? carried from the appliance track.
- **Region/capacity reality:** GPU stockouts are common (Mumbai was fully stocked-out on 2026-06-20); scale-to-zero + multi-region/flexible-zone selection is part of why Cloud Run+GPU fits.

- TPU is explicitly **out of scope**; this ADR records *why* so it isn't relitigated.

Seed scope for the "Cloud Run + GPU serving" subproject (formalize post-archival)
1. Containerize `backend.worker` (indexer venv + `wasb` conda env; NVDEC-capable base image).
2. Cloud Run + GPU (L4) service: `infer` over a `gcs://` workspace; scale-to-zero.
3. Per-video cost ledger (on the `jobs` table) = the per-customer bill (M3).
4. Cost guard + concurrency caps (M4).
5. Edge auth (tenancy boundary) before public exposure.
6. NVDEC GPU-decode to kill the extraction bottleneck.
7. DPA / consent gate for non-owner uploads (productization, not compute).

ADR-013: Drop DigitalOcean for compute ? GCP is the SOLE compute stack (training + serving)

Date: 2026-06-20 · **Status:** RATIFIED (owner, in-session: "drop DO and document it? record it and bake it for all future work"). **Supersedes the DO=training split in ADR-011; refines ADR-010 + ADR-012.**

Decision

1. **DigitalOcean is DROPPED as a compute provider.** **GCP is the sole compute stack** for BOTH training and serving. (ADR-011's "DO=training, GCP=serving" split is retired ? GCP does both. The DECOUPLED scope decision in ADR-011 ? M0?M2 + M3.5 ? still stands.)
2. **Standing default for ALL future compute work = GCP** (a GPU VM today via `deploy/gcp/`; Cloud Run+GPU for serving per ADR-012). Do **not** re-evaluate DO/Paperspace for compute without a new ADR.
3. The `$200` DO credits, if used at all, go **only to non-compute** that DO bills cleanly (CPU droplets / a site host / Spaces storage / bandwidth) ? **never GPU**.

Rationale (researched 2026-06-20, against DO's own docs)

- **DO GPU is neither enabled on the account nor a clean free-credits path.** The DO API `/v2/sizes` returns **0 GPU sizes** (GPU Droplets not enabled); GPU Droplets are **North-America-only** (NYC2/TOR1/ATL1/RIC1/AMS3 ? no India/Singapore); and DO's own docs state that for **newer customers DO charges the team's primary payment method** when GPU Droplets are created (anti-abuse) ? so GPU spend hits the **card**, not the promo credit. The ADR-011 premise ("$200 credits fund DO training") does **not** hold.
- **Paperspace (now DO Gradient) free GPU is subscription-gated** (a paid "G*"/Growth plan; high-end A100/H100 needs the $39/mo Growth plan), and Paperspace bills **separately** from DO Droplet credits ? the $200 credit doesn't fund it either.
- **GCP works today:** quota granted (`GPUS_ALL_REGIONS=1`), bucket + SA + `$100` budget provisioned, and M1/M2/M3.5 ran on an L4 for ~$1?2/run with no subscription. Real money, but trivial per run ? and **one** portable PyTorch+CUDA stack (local ? GPU VM ? Cloud Run), no second provider to maintain.

Consequences

- **Compute provisioning = `deploy/gcp/` only.** `SETUP_NEW_MACHINE.md` repointed to GCP; `deploy/digitalocean/README.md` marked **DEPRECATED**; `HOSTING_PLAN.md`'s "All-DigitalOcean" compute option retired.
- **`deploy/digitalocean/` scripts are KEPT** ? `bootstrap.sh` / `mount_scratch.sh` / `gpu_setup.sh` are **provider-agnostic** Linux-GPU setup the GCP runbook reuses; only the DO-specific *provisioning* (doctl droplet create, Spaces) is dropped. (Optional later cleanup: move them to `deploy/common/`.)
- The shared DO CPU droplet **`claude-do-rally-test`** (another session's testbed) is **owner-coordinated** cleanup, separate from this decision; the freshly-rotated **DO API token** can be retired once that droplet is gone.
- **Storage stays bucket-abstracted** (`gcs://` primary). The `s3` backend still speaks any S3-compatible store, so **DO Spaces remains a *possible* storage target** even though DO *compute* is dropped.
- **Capacity reality:** GCP GPU stockouts are common (asia-south1/Mumbai was fully stocked-out 2026-06-20 ? the M2 run landed an L4 in asia-southeast1/Singapore; weights still land in the asia-south1 bucket). Sweep zones/regions; scale-to-zero (ADR-012) is part of why Cloud Run+GPU fits.

- **DECOUPLED_COMPUTE** archives on M2 completion with a note that the DO leg was **dropped** (this ADR), not "delivered on DO".

ADR-014: Compute-decoupled rally serving ? one pure core, profile-selected compute + storage

Date: 2026-06-21 · **Status:** ? **RATIFIED (owner, 2026-06-21 ? M0 design sign-off; build M1+).** **Refines ADR-011 + ADR-012** (it *realizes* their deferred M3/M4; it does **not** supersede them); **builds on ADR-013** (GCP sole compute stack). Full design: `docs/COMPUTE_DECOUPLED_SERVING/`.

Evolution / lineage (trace the idea)
`ADR-005` (permissive licensing ? engine + image stay MIT/Apache/BSD) · `ADR-008` (train==serve featurizer single-sourcing ? the parity discipline this extends to local?cloud) · `ADR-009` (**local-first delivery** ? origin of the truly-local profile) · `ADR-010` (DECOUPLED DO?GCP) · `ADR-011` (**deferred M3 serving + M4 billing**; overrode "rent nothing / not a service"; §06 gates = DPA/consent, collector `md5`, WASB-C3) · `ADR-012` (**Cloud Run+GPU = serving model**; NVDEC lever) · `ADR-013` (**GCP sole compute stack**) · **? ADR-014** (this). Non-ADR: realizes `PLATFORM_ARCHITECTURE.md`'s `ComputeBackend`/`StorageBackend` registries; the truly-local profile = `VIDEO_LOCALITY_MODEL.md` L0 + `DESKTOP_APP_PLAN.md` `LocalDesktop`; informed by the archived `DEPLOY_REPORT_GCP_2026-06-20` (cross-GPU sub-pixel parity finding).

Decision

1. The rally engine is a **PURE function of (input bytes + config) ? outputs**, in three deterministic stages ? **DETECT** (`DetectorRunner.run_predict`?CSV), **WINDOW** (`trajectory_to_action_windows`, CPU-pure), **STITCH** (`VideoCompiler.compile_highlights`, ffmpeg) ? that are **NEVER branched on the deployment profile**.
2. A **`ComputeBackend` / deployment profile** decides **WHERE** the core runs (local process vs a Cloud Run GPU job); a **`StorageBackend`** decides **WHERE I/O comes from** (local FS / USB / mounted Google Drive / bucket). **Both are selected by ONE startup/setup config** (`deployment.profile` + `compute_target`), realizing the PLATFORM_ARCHITECTURE `local|hosted|byo_worker` registry.
3. Ship at least two profiles: **truly-local** (local GPU + **in-process stitch**, no cloud) and **cloud-serving** (a Cloud Run GPU job runs detect **AND** stitch), plus **cpu-mock** for CI and a reserved, **deferred** `byo_worker` hybrid.
4. In cloud-serving, **STITCH runs on Cloud Run** (co-located with detect) so its CPU/wall-time + egress are **metered** into a per-job cost ledger alongside `gpu_active_seconds` ? stitching is compute-intensive and its cost must be accounted for. (Truly-local stitch is intentionally **un**metered.)
5. **Parity** is enforced at **WINDOW + STITCH** (byte-exact for a fixed trajectory); **DETECT** is byte-exact only **same-GPU** ? cross-GPU is sub-pixel-different (deploy report 2026-06-20), so it is judged at the **rally/window level**, never CSV bytes.
6. Land **smallest-safe-first, one PR per milestone (M0?M10)**, **DEFAULT-OFF** where a change touches core callers (e.g. the configurable output base), each guarded by golden-fixtures + a **profile-parity test**; default flips ship a **Launch Report**.

Rationale

- The seams already exist: `_resolve_runner` is the single compute seam, the `StorageBackend` ABC + 4 backends + the worker fetch/put are built, and the stub detector (`replay_csv`) gives a **$0 deterministic** core. Realizing a profile system is **composition, not a rewrite**.
- Separating **WHERE** (ComputeBackend, coarse, per-job) from **HOW** (DeviceContext, fine, per-GPU) from **WHAT-bytes** (StorageBackend) keeps each axis independently testable and avoids the main trap ? conflating them.
- **Metering stitch on Cloud Run** is the only way the owner's real serving cost (GPU + CPU re-encode + egress) is attributable per job; stitch-on-laptop hides it.
- Aligns with ADR-012 (GPU-native) + ADR-013 (GCP sole stack) + the 2026-06-20 deploy report; the truly-local profile preserves the ADR-009 privacy/offline self-host story.

Consequences

- **Positive:** one core, provably identical across modes; CI proves the decoupling for **$0** (stub + storage fakes + a profile-parity test); new compute targets (byo_worker,

SkyPilot, Modal) slot into the enum without core changes; **cost becomes visible per job** (incl. stitch).
712   - **Cost/new surface:** a new `deployment`+`compute_target` config surface, a Cloud Run **dispatch shim**, a **containerized GPU+ffmpeg worker image**, a **cost-ledger migration**, and **edge auth** must all be built + maintained. Cross-GPU detection is **not** byte-identical ? parity asserted at the window level. The `byo_worker` tier carries a real ops/support burden (deferred, power-user only).
713   - **Risk-managed:** every core-touching change is default-OFF + golden/parity-gated + Launch-Reported; real-GPU runs are **owner-side** with a budget guard and a posterity DEPLOY_REPORT (`docs/COMPUTE_DECOUPLED_SERVING/runlogs/`).
714   - **Owner gates ? the 5 gating ones LOCKED 2026-06-21** (project doc §2 **D7?D12**): D7 cold-start = **scale-to-zero**; D8 pricebook = **versioned DB table, USD-internal/INR-display**; D9 edge-auth = **Cloudflare Access** (verify Cf-Access JWT, lock ingress); D10 = **free-tier data-for-reels trade** (adults-only, derived-data, counsel-gated to M9); D11 quality-floor = **Gemini-ON until the owned model clears the floor** + D12 the **data-flywheel harness** (capture?store?eval?goldenize). Plus **3 more locked at M0 sign-off (2026-06-21, §2 D13?D15):** **D13** profile is user-switchable anytime (advertised feature; equivalent output, not bit-identical cross-GPU); **D14 NORTHSTAR** = mobile/cloud-native point-at-Google-Drive ? reel back in Drive next to the source (implies a new **`gdrive-api` OAuth** backend = M12; resolves the gdrive-in-cloud question); **D15** auto-detect SUGGESTS, the user CONFIRMS ? execution (esp. cloud spend) is never a surprise. Still open (non-gating): stitch co-located vs split; NVDEC vs libx264; the `<2GB` cap; the dir name.

13. user (2026-07-08T05:45:54.102Z)

docs\CODE_MAP.md:22:Top-level tree and the **placement rule** for each kind of file. When unsure, prefer an existing
docs\CODE_MAP.md:162:[Omitted long matching line]
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:156:- Shallow section replacement behavior.
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:361: - `test_config_shallow_section_replacement`: verifies loader's shallow {**defaults, **file_data} (whole section from file replaces defaults section; no deep merge of sub-keys).
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md:150: - Restored the strong original `test_get_returns_dict_for_nested_sections_legacy_compat` (exact name + all legacy dict/nested/scalar/leaf-key/ default/fallback guards that the weaker shim replacement had dropped).
docs\HOW_RALLY_DETECTION_WORKS.md:32:From the per-frame `(x, y)` we compute the shuttle's **velocity** (displacement between frames).
docs\COMMERCIALIZATION.md:63:| **torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen replacement; person = COCO class (see `backend/pipeline/detectors/person_detector.py` `_PERSON_CLASS_ID`). |
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:52: wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*
docs\archives\roora-vendor\ROORA_LABELING_GUIDELINES.md:192:**Boundary placement is the non-negotiable part:**
docs\archives\roora-vendor\ROORA_LABELING_GUIDELINES.md:342:- `forced_error` = the opponent's shot CAUSED the miss (pace, placement, deception; the player barely
docs\COMPUTE_DECOUPLED_SERVING\README.md:90:### 4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
docs\COMPUTE_DECOUPLED_SERVING\README.md:140:2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
docs\KHELSUTRA_PRODUCT_OVERVIEW.md:161:- It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds insight.
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:7:**Key outcomes from #166 (merged c558372):** Hardening loop (T1-T5 + finale #165) COMPLETE and archived. NEXT_STEPS reframed with explicit P3 start signal (harness LOVO litmus for person features) + full prioritized candidate list. Gate A wired (opt-in, default-OFF safety per served regression litmus). Loop scheduler (019ec6d07da5) deleted; no replacement created. Handoff and archived audit updated terminal. Master also includes #167 (packaging).
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:21:[Omitted long matching line]
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:45:[Omitted long matching line]
docs\R2_EVAL_METRIC_DESIGN.md:156:- **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU (augment). The *gate*

```
docs\R2_EVAL_METRIC_DESIGN.md:188:- **Gate: AUGMENT now, then ablation-gate the eventual
replacement, then deprecate-with-docs ? ADOPTED.**
docs\PER_VIDEO_WORKSPACE.md:6:[Omitted long matching line]
docs\ROADMAP.md:32:pipeline objectively and (2) train an offline replacement for the paid Gemini
docs\RALLY_DETECTION_QUALITY_REPORT.md:288:aggressively (so the gate trims **real** rallies
instead). The honest, served replacement is **W2**
docs\STORAGE_SHARING_MODEL.md:185:13. On-device milestone placement: academy pilot boxes first,
consumer app after the WSL-free
docs\archives\past_projects\AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md:216:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md:369:[Omitted long matching line]
docs\TRACKNET_WSL_SETUP.md:67:  "velocity_threshold": 0.02    // normalised shuttle
displacement/sec to count as "in flight"
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_validation-
latency_2026-07-07.md:135:[Omitted long matching line]
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:46:| **D5** |
**Follow the CODE_MAP placement rules.** No new top-level files without a placement-rule match.
Tools go in `tools/`, standalone run-drivers in `scripts/`, eval library code in
`backend/eval/`. | `docs/CODE_MAP.md` §0 | Keeps the repo navigable as it evolves. |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:266:-
**CODE_MAP.md** `docs/CODE_MAP.md` ? placement rules, gotchas, and the cross-cutting footguns
list.
docs\archives\past_projects\DECOUPLED_COMPUTE\07-COMPUTE-ABSTRACTION.md:92:    def to(self,
model_or_tensor): ...           # placement (cuda/mps/cpu)
```

14. user (2026-07-08T05:45:55.212Z)

```
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_validation-
latency_2026-07-07.md:135:[Omitted long matching line]
```

15. user (2026-07-08T05:46:05.410Z)

```
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_alpha-
all1v1_2026-07-06.md:64:Rough (M8 ledger not landed): Cloud Run L4 `gpu_active` ? 1,800 + 2,412
+ 956 + 1,302 ? **6,470 s ? 1.8 L4-hours**, plus the diag VM ? 1 L4-hour (g2-standard-4). At
~US$0.85 / L4-hour on-demand ? **? US$2.4** for the whole diagnostic session. The proxy path is
**~50 % fewer detection GPU-seconds** per clip going forward.
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_alpha-
all1v1_2026-07-06.md:66:## 8. Teardown / final state ($0 audit)
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_alpha-
all1v1_2026-07-06.md:68:- Diag VM **deleted** (never stopped): `gcloud compute instances list` ?
**0 items**; `gcloud compute disks list` ? **0 items** (no orphan boot disk). ? $0.
docs\COMPUTE_DECOUPLED_SERVING\runlogs\README.md:26:8. **Teardown / final state** ? for cloud,
the `$0` audit (`gcloud compute instances list` + `disks list` empty); **DELETE, never just
stop** ([[gcp-vm-always-delete]]).
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_cp-rebuild_2026-07-05.md:62:
after hashing. Cost of the encounter here: one L4 re-detection (~US$0.4), harvested as the
docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_truly-local_TEMPLATE.md:41:## 4. Parity sanity
(optional, $0)
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_validation-
latency_2026-07-07.md:100:- 2 L4 worker execs (Run A 157.7 s + Run B 109.6 s ? 268 s of L4 time)
? **~$0.5?0.7**. Off-box A/B: **$0** (local). No VM, no image build. GCS read for the 3
downloaded clips (~5.1 GiB) is intra-project.
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_validation-
latency_2026-07-07.md:137:**Cost (heuristic):** one L4 detect exec (`rally-worker-64bhj`, ~222
s) ? **$0.15?0.55** (rate-dependent, dominant) + control-plane CPU for validation+stitch (~960 s
@ 2 vCPU) ? **~$0.05** + reel egress (581 MB, if downloaded to internet) ? **~$0.07** +
storage/ops **< $0.02** + Gemini **$0** (skip_ai_handoff). **? $0.3?0.7 total**, ~one L4 exec as
projected ? but the stitch's real cost is **concurrency** (~16 min of control-plane heavy CPU),
not dollars.
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:3:> **Date:**
2026-07-04 · **Owner-authorized** GCP spend (D17, $300 envelope) · deleted every
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:4:> resource
after (`[[gcp-vm-always-delete]]`; §5 residual audit = $0 of ours).
```

docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:52:- Builds: 4 × ~$0.16 (cu113 base reuse, cu118, batch) ? $0.5
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:53:- Runs: baseline + cu118 + batch64 + batch128(OOM, short) + probes ? ~$2?3 of L4 time
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:54:- Idle: **$0** (scale-to-zero; GPU bills only during execution)
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:55:- **Total well under the $300 / D17 envelope.**
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:57:## 5. Teardown ? residual audit ($0 of ours)
docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_2026-07-04.md:70:No GPU/compute of ours remains; idle cost = $0.
docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_cloud-serving_exec-status-fastfail_2026-07-07.md:70:### A ? classifier vs live history (read-only, $0)
docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_cloud-serving_exec-status-fastfail_2026-07-07.md:132:- **A:** $0 (read-only `list_executions` + `get_execution`).
docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_cloud-serving_exec-status-fastfail_2026-07-07.md:136:- Total: well under $1.

16. user (2026-07-08T05:46:07.116Z)

01-EXISTING-PLANS-AND-GAP.md
02-COMPUTE-MAP-AND-CONTRACT.md
03-PLATFORM-OPTIONS.md
04-BUILD-AND-TEST-PLAN.md
05-COST-ATTRIBUTION.md
06-RISKS-AND-OPEN-QUESTIONS.md
07-COMPUTE-ABSTRACTION.md
DEPLOY_REPORT_GCP_2026-06-20.md
HANDOFF.md
README.md

17. user (2026-07-08T05:46:15.975Z)

05 · Cost attribution ? per-video inference & per-run training billing

Status: PROPOSAL (net-new design) · **Date:** 2026-06-20

> **Side goal:** a non-owner user runs a job and gets the **exact billing for that video** (inference), and
> training cost can be **segmented similarly** (per run, optionally amortized per contributed video). This
> is wholly new ? the `jobs` table has **no cost columns** today ([`database.py:188`](../../backend/infrastructure/database.py)).
> The good news: a job is already the unit of work, and serverless platforms already meter per invocation ?
> so per-video billing is mostly *bookkeeping on top of seams that exist.*

1. The core idea

A job is the billing unit. An `infer` job is **1:1 with a video** (keyed by `video_id`/MD5), so the
job's cost *is* the per-video bill. A `train` job is **1:1 with a training run**, so its cost is the run
cost ? which we then optionally **split across the videos that fed it**.

```
cost_usd(job) = ?_resource  usage(job, resource) × rate(resource, pricebook_version)
```

Resources to meter: **GPU-seconds** (the big one), CPU-seconds, wall-seconds, **egress GB** (reel
downloads ? the line `HOSTING_PLAN.md` flags), storage GB-hours, and **external-API calls** (Gemini
tokens/calls ? the provider already returns metrics: `analyze_video ? (segments,

metrics)`).

2. Metering mechanics (cheapest accurate source first)

Source of GPU-seconds	When	Accuracy
Serverless per-invocation report (RunPod Serverless / Cloud Run / Modal)	M3 inference endpoint	**Best** ? the platform bills exact GPU-seconds per job; that number *is* the bill. **This is the strongest argument for a serverless `infer` topology** ([03](03-PLATFORM-OPTIONS.md)).
Worker-measured `gpu_active_seconds` (timestamp around the GPU section, × the known `gpu_type` rate)	DO Droplet / VM where there's no serverless meter	Good ? separate `gpu_active_seconds` from `wall_seconds` so you don't bill the user for the bucket-sync/upload time.
Provider billing API reconciliation (monthly)	all	Truth source ? reconcile the sum of job costs against the invoice; carry a `reconciliation_factor`.

Egress is counted from bytes downloaded (R2 = zero egress ? simplest). Gemini cost comes from the metrics
the provider already returns.

3. Schema delta (extends the existing `jobs` table; no new datastore)

Add via the existing `_add_column_idempotent` / `PRAGMA user_version` migration ladder:

```sql
ALTER TABLE jobs ADD COLUMN owner_id        TEXT;     -- who pays (the non-owner user / tenant)
ALTER TABLE jobs ADD COLUMN compute_backend TEXT;     -- runpod_serverless | gcp_cloudrun | do_droplet ?
ALTER TABLE jobs ADD COLUMN gpu_type        TEXT;     -- L4 | A4000 | T4 ?
ALTER TABLE jobs ADD COLUMN gpu_active_seconds REAL;
ALTER TABLE jobs ADD COLUMN wall_seconds       REAL;
ALTER TABLE jobs ADD COLUMN bytes_out          INTEGER;   -- egress
ALTER TABLE jobs ADD COLUMN gemini_cost_usd     REAL;
ALTER TABLE jobs ADD COLUMN pricebook_version  TEXT;
ALTER TABLE jobs ADD COLUMN cost_usd            REAL;     -- computed total
ALTER TABLE jobs ADD COLUMN cost_breakdown_json TEXT;     -- {gpu: .., egress: .., api: ..}

-- price book (versioned; rates change, and differ per provider/gpu_type)
CREATE TABLE IF NOT EXISTS pricebook (
  version TEXT, resource TEXT, gpu_type TEXT, unit_price_usd REAL, currency TEXT, effective_at TIMESTAMP
);

-- training-run cost attribution across contributing videos
CREATE TABLE IF NOT EXISTS training_contributions (
  training_job_id TEXT, video_id TEXT, weight REAL   -- weight = clip-seconds share (or 1/N)
);
```

The **`pricebook` is seeded from the same provider-grounded GPU set as the device profiles**
([07](07-COMPUTE-ABSTRACTION.md) §5) ? DO/Paperspace now (A4000 / L40S / A100), GCP later (T4 / L4 / A100) ?
so **one `{provider, gpu_type}` table carries both the $/hr rate (here) and the device-tuning knobs (there)**,
keyed off the `gpu_type` that `run_telemetry`'s `nvidia-smi` probe already yields.

`owner_id` is the per-tenant scoping the multi-tenant model ([`PLATFORM_ARCHITECTURE.md`](../PLATFORM_ARCHITECTURE.md) §4)
will need anyway ? adding it now keeps "a non-owner user is billed for their jobs" a first-class property.

4. Per-video inference billing

1. An `infer` job's `cost_usd` + `cost_breakdown_json` **is** the per-video bill (job is 1:1 with the video).
2. Expose `GET /api/jobs/{id}/cost` and a per-video cost history (`GET /api/videos/{video_id}/cost`), scoped by `owner_id`.
3. Optional resale markup: a `margin` config ? `price_to_user = cost_usd × (1 + margin)`; store both so the user sees a transparent breakdown and you keep your margin auditable.

Example breakdown (illustrative): 3-min L4 inference + a 40 MB reel download + one Gemini call ? `gpu: 180 s × $0.00022/s = $0.040` + `egress: 0.04 GB × $0 (R2) = $0` + `gemini: ~$0.05` ? **~$0.09/video**.

5. Per-training cost segmentation

1. A `train` job's `cost_usd` is the **run cost** (e.g. Gen-0 ? a few cents; a future VLM run would be $-tens).
2. To "segment similarly," attribute the run cost across the N videos/labels in that run via `training_contributions` ? **equal split (`1/N`)** or **weighted by clip-seconds contributed**. This yields *"cost per video contributed to training,"* the symmetric analog of the per-video inference bill.
3. Record `training_job_id` on the produced weight bundle so a served model's lineage carries its training cost (useful for amortizing training into a per-inference price later, if you choose).

6. Trust & reconciliation

- **Transparency:** always show the user the `cost_breakdown_json` + `pricebook_version`, not just a total.
- **Reconciliation:** metered cost ? provider invoice but not exact (overhead, rounding, idle). Reconcile monthly: `reconciliation_factor = invoice_total / ? metered_cost`; surface drift if it exceeds a few %.
- **Spend guard:** a per-`owner_id` rate limit + a global hard cap (the same cap that protects the $200) means a leaked API key can't run up someone else's bill or drain the budget.

7. Why this is cheap to build

It rides entirely on seams that exist: the **`jobs` table** (already the unit of work + has `compute_backend` in the design), the **per-video output layout**, the **`tenant_id`** the platform doc already plans, and the **ComputeBackend** (which knows `gpu_type` + rate). The only genuinely new pieces are the **pricebook**, the **usage capture in the worker**, and the **attribution table** ? all small, and none requiring a new datastore.

18. user (2026-07-08T05:46:26.294Z)

```
backend\api\job_worker.py:201:    (= {gpu_seconds, wall_seconds, egress_gb, gemini_usd, gpu_type}) × the versioned pricebook, and
backend\api\job_worker.py:205:    ``billing.enabled=True`` AND a real calibrated pricebook, every recorded cost is **0** until a
backend\api\job_worker.py:206:    FOLLOW-UP wires the worker to emit usage (gpu/wall/egress/gemini). M8 lands the ledger + pricebook
```

```
backend\api\job_worker.py:208:    With the placeholder pricebook the cost is 0 regardless.
backend\api\job_worker.py:221:            pricebook_version=billing_cfg.pricebook_version,
backend\api\job_worker.py:228:            "Recorded cost for job %s (pricebook=%s).",
backend\api\job_worker.py:230:            billing_cfg.pricebook_version,
backend\billing.py:3:``cost_usd(job) = ?_resource usage(job, resource) × rate(resource, pricebook_version)``
backend\billing.py:6:(the Bangalore market). The pricebook is a versioned DB table (``backend/infrastructure/database.py``);
backend\billing.py:7:this module is the pure arithmetic over a resolved ``{resource: unit_price_usd}`` rates dict, so it
backend\billing.py:10:**Default-OFF:** with the seeded ``placeholder-v0`` pricebook (every rate ``0.0``) every cost is
backend\billing.py:11:``0.0`` ? nothing user-facing changes until the owner inserts a real, calibrated pricebook version.
backend\billing.py:18:# Metered resource keys ? shared by the usage dict, the pricebook rows, and the breakdown.
backend\billing.py:28:# Resources whose cost = amount × unit_price (GEMINI_USD is excluded ? it is already a USD amount).
backend\billing.py:38:    key to its USD unit price (already resolved for this pricebook version + gpu_type by the DB
backend\billing.py:40:    Missing amounts/rates default to ``0.0`` ? with the placeholder pricebook the total is ``0.0``.
backend\config\models.py:491:    ``pricebook`` DB table and **displayed in INR** at ``fx_inr_per_usd``. The default pricebook
backend\config\models.py:492:    version is the seeded ``placeholder-v0`` (every rate ``0.0`` ? cost ``0``) until the owner
backend\config\models.py:493:    inserts a real, calibrated version and points ``pricebook_version`` at it.
backend\config\models.py:495:    ? Even with ``enabled=True`` + a real pricebook, recorded cost stays **0** until a follow-up
backend\config\models.py:497:    pricebook + the recording seam; usage emission (the metering hooks) is a separate step."""
backend\config\models.py:500:    pricebook_version: str = (
backend\config\models.py:501:        "placeholder-v0"  # the seeded $0 version; owner flips to a real one
backend\main.py:981:    null until billing records it (default-OFF / the placeholder pricebook ? 0)."""
backend\infrastructure\database_jobs.py:316:    def get_pricebook_rates(

[Showing results with pagination = limit: 25]
```

19. user (2026-07-08T05:46:27.266Z)

```
# DEPLOY REPORT ? cloud-serving, GPU-resident NVDEC decode rollout (#506/#507), 2026-07-07

> Durable record of shipping the **GPU-resident WASB decode** (`decode_backend=nvdec_resident`, issue #506 / PR #507) to the live `cloud-serving` path: rebuild + cut over the `rally-worker` L4 Cloud Run Job image (torch 1.11 ? **2.6.0+cu124 + PyNvVideoCodec 2.1.0 + baked NVDEC libs**), smoke the cpu + nvdec paths on the real L4, then **enable nvdec via a reversible Job env var** and canary it. Project `gen-lang-client-0306026826`, region `asia-southeast1`. Honest-limits in §6.

## 1. TL;DR

| Stage | Result |
|---|---|
| Merge | ? PR #507 squash-merged to master (`44a2f47`); full local gate + CI green |
| Build image | ? `rally-worker:latest` ? `sha256:465a9250?` (Cloud Build 12m7s) |
| Job cutover | ? `jobs update --image ?:latest` (prior digest `sha256:3bfefaf6?` = rollback) |
| Phase A ? cpu smoke (torch 2.6) | ? 180s clip: `torch 2.6.0+cu124 cuda True`, 17 windows, WASB 201.9s |
| Phase B ? nvdec smoke (fast) | ? 180s clip: 19 windows, WASB **143.7s (~1.4× vs cpu)**, 2934 vis ? cpu 2920 |
```

Phase B ? nvdec smoke (**hard GX010128**)	? **26479/45298 visible ? VM 26481** (?0.004%), 85 windows, WASB 1138s (~40 fps)
Enable	? `WASB_DECODE_BACKEND=nvdec_resident` set on the Job (reversible)
Canary (env-driven)	? 180s clip, **no** decode `--set` ? env var selected nvdec: 19 windows, 2934 vis, WASB 151.4s (== explicit-`--set` nvdec)

Bottom line: the GPU-resident decode path runs on the real Cloud Run L4 and **reproduces the validated VM trajectory** (26479 ? 26481 visible on GX010128), so the golden-gate F1 (0.574 ? baseline 0.582 @ SERVED) transfers to serving. nvdec is enabled via a Job env var with an instant rollback.

2. Resources

- **Worker:** `rally-worker` Cloud Run **Job**, image `asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest` (`sha256:465a9250?`), 8 vCPU / 32 GiB / 1× NVIDIA L4, task-timeout 3600 s, GCSFuse mount of `gs://khelsutra-rally-serving`, `WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar`, **`WASB_DECODE_BACKEND=nvdec_resident`** (the enablement lever).
- **Image change (PR #507):** WASB conda env py3.8+torch1.11 ? **py3.10 + torch 2.6.0+cu124 + PyNvVideoCodec==2.1.0**; `libnvcuvid`/`libnvidia-encode` (from driver 580.159.03) baked in a multi-stage layer. cpu decode path byte-unchanged; nvdec is a new, cache-isolated path.
- **No VM used** for this deploy (the image build is Cloud Build; smokes run on the existing Job).

3. Exact commands

```bash
# Build + push (from master @ 44a2f47)
gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project gen-lang-client-0306026826 .
# Cut the Job over to the new image (re-resolves :latest to the new digest)
gcloud run jobs update rally-worker --region asia-southeast1 --image ?/rally/rally-worker:latest
# Smoke (direct Job execute; skip_ai = secret-free; --set decode_backend per leg). ^@@^ arg delimiter.
gcloud run jobs execute rally-worker --region asia-southeast1 --async \
  --args='^@@^infer@@--video-uri=gs://khelsutra-rally-corpus/videos/<clip>@@--out-uri=gs://khelsutra-rally-serving/rel506/<label>@@--segmenter=wasb_hybrid@@--set=deployment.compute_target=local_gpu@@--set=indexing.detector_impl=native@@--set=indexing.skip_ai_handoff=true@@--set=indexing.wasb.batch_size=64@@--set=indexing.wasb.prefetch_batches=4@@--set=indexing.wasb.timeout_sec=0@@--set=indexing.wasb.decode_backend=<cpu|nvdec_resident>'
# ENABLE (reversible): the native runner reads WASB_DECODE_BACKEND first, so this flips all dispatches
gcloud run jobs update rally-worker --region asia-southeast1 --update-env-vars WASB_DECODE_BACKEND=nvdec_resident
# ROLLBACK options
gcloud run jobs update rally-worker --region asia-southeast1 --remove-env-vars WASB_DECODE_BACKEND   # nvdec ? cpu (same image)
gcloud run jobs update rally-worker --region asia-southeast1 --image ?/rally/rally-worker@sha256:3bfefaf6?  # full revert to the torch-1.11 image
```

4. Run results (executions)

Exec	Clip	decode	via	visible / frames	windows	WASB s	status
rally-worker-p2wd6	mahadevpura_smoke_180s	cpu	`--set`	2920 / 5395	17	201.9	?
rally-worker-clkwc	mahadevpura_smoke_180s	nvdec_resident	`--set`	2934 / 5395	19	143.7	?
rally-worker-6htsw	**GX010128**	nvdec_resident	`--set`	**26479 / 45298**	85	1138.0	?
rally-worker-cnzxw	mahadevpura_smoke_180s	nvdec (via **env**)	Job env var	2934 / 5395	19	151.4	?

50 - GX010128 nvdec on Cloud Run: **26479 visible vs the validated VM run's 26481** (? 2 frames / 45298 = 0.004%) ? the serving trajectory matches the golden-gate trajectory, so its **F1 0.574 @ SERVED (? baseline 0.582)** holds in prod.
51 - Fast-clip nvdec vs cpu (same tuning batch 64 / prefetch 4): WASB **143.7 s vs 201.9 s (~1.4×)**, near-identical trajectory (? visible 0.3%).
52
53 ## 5. Rollout state + rollback
54 - **Enabled** on `rally-worker` via `WASB_DECODE_BACKEND=nvdec_resident`. All future dispatches use the GPU-resident path.
55 - **Rollback (instant, same image):** `--remove-env-vars WASB_DECODE_BACKEND` ? back to the (torch-2.6) cpu path. **Full revert:** `--image sha256:3bfefaf6?` ? the prior torch-1.11 image.
56 - **Durability (landed):** PR #508 set `indexing.wasb.decode_backend: nvdec_resident` in the `cloud-serving` preset. But the worker Job has **no `DEPLOYMENT_PROFILE`**, so it never applies that preset ? a follow-up (`chore/506-release-hardening`) makes the control plane **forward** the resolved `indexing.wasb.{decode_backend,batch_size,prefetch_batches}` to the worker (`orchestration.py`), so the preset is the true source of truth on the dispatch path. Until the control plane is redeployed with that fix, the `WASB_DECODE_BACKEND` env var above is what keeps nvdec live (and remains a valid manual override afterwards). ? The env var also revealed that the preset's `batch_size=64/prefetch_batches=4` were **never reaching the worker** pre-fix (dispatch forwarded only `skip_ai_handoff`) ? the forwarding fix closes that latent gap too.
57 - **Update (later 2026-07-07) ? forwarding is now LIVE.** Rebuilt the image from master `66756ad` (= #507+#508+#509) ? **`sha256:09d2edf7`** and redeployed BOTH surfaces to it: control-plane service rev **`rally-control-plane-00017-qbd`** (100% traffic, booted healthy ? uvicorn up, cloud-serving profile ACTIVE, edge-auth on) and the `rally-worker` Job. So the dispatch forwarding + the preset default now govern the worker from the image; the `WASB_DECODE_BACKEND` env var is retained as a redundant belt-and-suspenders override (both say nvdec). ? The FIRST control-plane deploy attempt used a **stale image** `sha256:465a9250?` (built from `44a2f47`, pre-#508/#509 ? no forwarding); it became rev 00016 and was superseded by 00017. Not live-dispatch-verified end-to-end (the `/api/process` path is behind Cloudflare Access ? no CF JWT this session); forwarding is deployed + unit-tested. **Control-plane rollback:** `gcloud run services update-traffic rally-control-plane --region asia-southeast1 --to-revisions rally-control-plane-00015-kp9=100` (last pre-nvdec revision).
58
59 ## 6. Honest limits / not-verified
60 - Smokes ran with `skip_ai_handoff=true` (secret-free) so the persisted "rallies" are the local candidate windows, not Gemini-refined ? this validates the **decode + detection + windowing** path, not the AI handoff (unchanged by this PR).
61 - F1 parity is asserted by the **offline golden gate** (10 clips @ SERVED) + the **trajectory reproduction** above (Cloud Run 26479 ? VM 26481 visible on GX010128); no on-Cloud-Run re-scoring against labels was run (the trajectory match makes it redundant).
62 - The control-plane **service** was redeployed to `sha256:09d2edf7` (see §5 Update), so the API path now runs the merged code (forwarding + persisted `decode_backend` metadata). The worker Job is the release-critical surface (it's the one that decodes) and was validated first.
63 - NVDEC driver-ABI: the baked `libnvcuvid 580.159.03` works on the current Cloud Run L4 host driver (proven by the successful decode). A future Cloud Run host-driver change could in principle skew ? re-smoke on any base-image/driver bump (the Dockerfile comment flags this).
64

20. user (2026-07-08T05:46:37.849Z)

1 # DEPLOY REPORT ? control-plane durable rebuild (CP-rebuild P9), 2026-07-05
2
3 > The §4.8/§9 execution record for `docs/CONTROL_PLANE_DURABLE_REBUILD.md` (spec #487, owner-confirmed).
4 > Everything here ran live against project `gen-lang-client-0306026826`, region `asia-southeast1`,
5 > on 2026-07-05 (14:10?14:5x UTC). Honest-limits in §6: what this run does NOT prove.
6
7 ## 1. What shipped in the image
8
9 Master `257f469` ? the full P-row sweep, all merged on verified LGTMs the same day:
10 P1 #489 (generator pins + 8Gi) · P2 #490 (GCS-streamed upload staging) · P4 #492 (submit-time

MD5 + cache-reuse) · P5a #491 (poll budget 3900 s + capabilities caps) · P6 #493 (cloud-poll heartbeat) · P3 #494 (restart-safe recovery + read-path re-hydrate) · P8 #495 (SPA re-vendored at khelsutra `90d1d90` = #153/#155/#158/#162 baked; platform-stable `ui_version` hash). khelsutra side: #162 (60-min SPA ceilings) on khelsutra master `90d1d90`.

2. Build ? pin ? deploy (§4.8 steps 1?4)

Step	Command / object	Result
Image build	`gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml` from a clean worktree at `257f469`	build `47877373` **SUCCESS in 7m33s** ? digest `sha256:3bfefaf6ad4a?7671b964` (moved from the prior live `31475f27?`)
Worker re-pin	`gcloud run jobs update rally-worker --image ?/rally-worker:latest`	job resolves the new digest
CP deploy (auth ON)	`gcloud run services replace service-auth-on.yaml` (generated by `deploy/cloudrun/gen_service_yaml.py --edge-auth on` from the clean clone)	rev **`rally-control-plane-00012-wxk`** · **8Gi** / 2 vCPU · imageDigest == `3bfefaf6?`
Pins in the baked config `[verified pre-deploy]`	base64-decoded the manifest's wrapper payload	`indexing.default_segmenter="wasb_hybrid"` + `security.max_part_mb=24` + `edge_auth_enabled=true` present ? **D5 works from a clean clone**
Upload-staging lifecycle	`gcloud storage buckets update --lifecycle-file ?`	`videos/uploads/**` age-2d Delete rule live (bucket previously had NO lifecycle policy)

3. Posture (§9, on the final auth-ON revision 00012)

- `GET /api/health` (identity token) ? **200** `{"status":"ok","sport":"badminton","default_segmenter":"wasb_hybrid"}` ? the segmenter pin is visible in the health payload itself.
- `POST /api/process` without a CF JWT ? **401** (edge-auth ON).
- `https://api.khelsutra.guru/api/health` ? **302** (Cloudflare Access intercept).

4. Verification battery (short auth-OFF window, rev `00013-rlc` ? the precedented A1-e2e pattern)

The auth-off deploy replaced the instance ? **fresh, empty SQLite: exactly the 2026-07-05 post-restart state** the spec's DoD demands be replayed.

Proof	What ran	Result
Restart replay (#471/#485 ? THE loss shape)	`POST /api/query {"video_id":"c2d45319?f55369"}` against the wiped DB	**29 play_segments returned** ? re-hydrated from `outputs/<md5>/` on demand (`metadata.source=cloud_run` rows). The July-5 morning failure shape now recovers by itself
Streamed upload staging (P2/#480)	2×768 KiB parts ? `/api/upload/init ? part/0 ? part/1 ? complete`	`path=gs://?/videos/uploads/7383?/p9-stream-proof.mp4`, `video_id=dc65ef78?`; **server-side object `md5Hash` (hex) == the streamed video_id EXACTLY**, `component_count` empty (non-composite) ? browser-staged uploads stay md5-addressable, so the instant cache path holds for the tester flow
Capabilities (P5a/#480 item 5)	`GET /api/capabilities`	`max_upload_mb=4096`, `max_part_mb=24`, `default_segmenter=wasb_hybrid`, `deployment.cloud_available=true`
Heartbeat (P6/#482 CP slice)	`GET /api/jobs/{id}` during the live detection	`progress = "segmenting (wasb_hybrid) on cloud GPU ? 5 min elapsed"` ? later ticks 6, 7? ? the silent 22-minute spinner is dead
Jobs queue (#481)	`GET /api/jobs?status=all`	lists the running job with its live progress
Fresh detection on the new image	the same submit dispatched `rally-worker-2mz8t` (see §5 for WHY it dispatched)	job ? **done, 48 segments**, `compute.path=cloud_run`, exec `?/rally-worker-2mz8t`, **wall 14m04s** (14:23:41?14:37:45Z). Clarifier: the bucket's `MahadevpuraSingles.MP4` hashes to `9154737?` ? a DIFFERENT file than the morning's 2.68 GB browser upload (`c2d45319?`/29 rallies); 48 matches the A1-e2e record for this original, and its `outputs/` were idempotently rewritten
Cache proof ? the literal §9 box (#484)	immediate re-submit of the same `gs://`

path (same instance, DB now warm) | **`done cached=true, path=cached, 48 segments` in 11 s**; `latestCreatedExecution` unchanged (`2mz8t`) ? **zero GPU on the re-submit** |

5. Findings (honest ? two real ones)

1. **Composite originals bypass the submit-time cache (documented P4 boundary, now measured).**
   `videos/MahadevpuraSingles.MP4` was bucket-uploaded as a **composite object** (`component_count=4`) ? GCS exposes **no `md5Hash`** ? `submit_time_video_id` returns None ? the instant-done path can't fire. By itself that would only degrade to the drain-tier DB check; combined with (2) it re-dispatched.
2. **Fresh-DB + composite-original ? the bucket cache is never consulted on the drain path.**
   Cloud Run replaced the rev-00013 instance between the replay query and the submit (`MANUAL_OR_CUSTOMER_MIN_INSTANCE` churn in the logs) ? the drain hashed on an empty DB, found no video row, and dispatched ? even though `outputs/<md5>/report.json` existed. The drain checks only the DB after hashing; the bucket tier lives on the submit path (md5-known) and the read path. **Residual gap, bounded:** it needs BOTH a composite/md5-less source AND an instance swap; browser-staged uploads (the tester flow) always carry `md5Hash` and take the instant path. Follow-up filed on #484: consult `cached_process_result` on the drain after hashing. Cost of the encounter here: one L4 re-detection (~US$0.4), harvested as the fresh-detection proof below.

6. What this run does NOT prove (stays on the owner's §1.1 CUJ)

- The **?2.5 GB browser upload** through Cloudflare Access on the real domain (the §9 headline memory-survival proof under load) ? the synthetic upload proves the staging PATH, not the 4 GiB-scale memory profile. The owner's official 5-minute CUJ covers it on this build.
- SPA-rendered UX (progress copy, medium picker, pickable rallies) ? API-level proofs only here; the vendored snapshot provenance (`ui_version.spa_commit=90d1d90`, content-hash guard in CI) is what ties the served UI to the verified khelsutra master.
- Concurrent-duplicate submit collapse (#481) was regression-proven in CI (12-thread test) but not re-driven live in this window.

7. Restore + final posture

`services replace service-auth-on.yaml` ? **rev `rally-control-plane-00014-7hk`** (edge-auth ON, 8Gi, same digest `3bfefaf6?`). Re-verified: health (identity token) **200**, `POST /api/process` sans CF JWT **401**. The `p9-stream-proof.mp4` staging object was deleted (the age-2d lifecycle rule would have collected it regardless). The auth-off battery jobs' DB rows died with the window's instance ? by design and now harmless: their results live in the bucket and any read re-hydrates on demand (§4 row 1 is exactly that proof).

8. Tracker updates riding this PR

- `docs/CONTROL_PLANE_DURABLE_REBUILD.md` §7: **P9 ? ?** (this runlog); status header

```
note.
- `docs/ALPHA_LAUNCH_READINESS.md`: DoD item 1 annotated ? the infra blocker behind the failed
  2026-07-05 owner CUJ is deployed-fixed (rev 00012-wxk); the official CUJ pass remains the
  owner's checkbox. A1 row cross-references this runlog.
- Issues: #480 + #471 + #485 closed with live evidence; #484 kept open (composite-gap follow-up);
  #482 stays open (worker-side heartbeat + per-job runlog).

```

21. user (2026-07-08T05:46:51.355Z)

```
# Control-Plane Durable Rebuild ? kill the upload OOM, ground state in the bucket, unblock DoD #1

> **Status:** `IN PROGRESS` ? **P9 EXECUTED 2026-07-05**: live = rev **00012-wxk** (8Gi, digest
> `3bfefaf6?`, edge-auth ON, pins live). §9 residual = the owner's §1.1 browser CUJ (three boxes);
> flip to `DONE` when it passes · **Owner:** avidullu · **Created:** 2026-07-05 ·
> **Last updated:** 2026-07-05
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`;
> pointer in memory `current-state` (+ `docs/NEXT_STEPS.md` once work starts post-confirmation).
> **Relation to existing docs:** executes the serving hardening that
> [ALPHA_LAUNCH_REVIEW_2026-07-04.md](ALPHA_LAUNCH_REVIEW_2026-07-04.md) §1.1 (the 5-minute browser
> CUJ ? tracker **DoD #1**) is blocked on; child of
> [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) (ADR-014); consolidates issues
> #471 #480 #482 #483 #484 #485 (plus the already-shipped #479-config, #481, and khelsutra
> #153/#154(#155)/#157(#158)).
> **Honesty note:** each claim is tagged `[verified]` (checked against code or live logs),
> `[researched]` (external platform fact ? re-verify during execution), or `[design]` (proposed).

---

## 0. TL;DR

The first real owner-driven browser CUJ (2026-07-05) proved the product works end-to-end ? 29 rallies
detected, pickable in the UI ? **when the control plane survives**. It also proved the deployed CP
does not survive real usage: a 2.68 GB browser upload staged into RAM-backed `/tmp` on a **4 GiB**
instance ? OOM ? restart ? ephemeral SQLite wiped ? the 29 rallies (already written to the bucket)
were never ingested ? empty UI, unrecoverable by reload `[verified]`. This spec consolidates the fixes
into **one image rebuild + redeploy**: ?8 GiB CP with upload staging **streamed to the bucket** (never
assembled in tmpfs) · **bucket-grounded durable state** (the DB becomes a cache that re-hydrates from
`outputs/<md5>/`) · **submit-time MD5** (makes re-runs instant AND restarts recoverable ? it is the
key that re-associates an orphaned job with its output) · longer, honest poll ceilings · the rev-00011
config pins **baked into the committed manifest generator** so a regenerate can never
```

silently drop
33 them · the current SPA (#153/#154/#158) re-vendored. The deploy step (**P9**) is owner-gated on
34 confirming this spec; the code rows (P1?P8) are each one small PR.
35
36 ## 1. Problem & goal
37
38 **What happened (2026-07-05, all diagnosed live ? full record in memory
39 [[alpha-cuj-testing-2026-07-05]]):**
40
41 1. 95 MB upload parts were 413'd by the Google Front End ? the service has no HTTP/2 annotation, so
42 ingress is HTTP/1 with a ~32 MiB per-request body cap (#480). Config rev 00011 set
43 `security.max_part_mb=24` and small uploads started working `[verified]`.
44 2. Cloud jobs ran `yolo_hybrid`, not the validated `wasb_hybrid` (#479) ? the engine default
45 (`backend/config/models.py:181`) was never pinned. Config rev 00011 pinned it `[verified]`.
46 3. **The killer:** a 2.68 GB upload assembled into the CP's RAM-backed `/tmp` on a 4 GiB instance ?
47 OOM ? restart 08:14Z ? the ephemeral SQLite **and the in-memory done-marker poll** were wiped
48 (#471). The worker finished (`done rc=0` 08:59:51Z) and wrote 29 rallies to the bucket, but no CP
49 was listening ? never ingested ? the UI showed nothing and reload could not recover it
50 `[verified]`. The reel was recovered manually from `outputs/<md5>/intervals.csv`.
51 4. Timeouts are dishonest for real footage: the SPA gives up at 10 min
52 (`web/src/hooks/useIngestJob.ts:18`, `web/src/api/client.ts:181` in the khelsutra repo) and the CP
53 poll budget is 30 min (`backend/infrastructure/execution_policy.py:54`), while a 6-min 1080p60 clip
54 takes WASB ~22 min on the L4 `[verified]`.
55 5. No MD5 reuse: the same clip was detected **3×** that day ? three paid L4 runs for one video
56 (#484) `[verified]`.
57 6. The gs:// re-run CUJ **PASSED in the UI** (29 rallies pickable) because nothing staged into tmpfs ?
58 the product works when the CP survives `[verified ? owner screenshot]`.
59
60 **Goal.** A tester can, from a browser at `https://api.khelsutra.guru` (edge-auth ON), upload a real

22. user (2026-07-08T05:46:52.836Z)

19:[Omitted long matching line]
20:[Omitted long matching line]
36:[Omitted long matching line]

23. user (2026-07-08T05:46:59.939Z)

1 - RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
2 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution

doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

- PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.

- RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.

- EVALUATION.md ? How we measure and improve quality.

- EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

- QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

- ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).

- PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

- GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

? Documentation status & governance

- [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE / ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit +

its folded-in **Remediation tracker** (P0?P28 all closed). **Archived 2026-07-02** once every remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444 corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior `CODE_CLEANUP_AUDIT_2026-06-24` + June-2026 hardening sweep (both archived).
15 - DOC_STATUS.md ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.
16
17 ## Product / platform plans
18 - ALPHA_LAUNCH_READINESS.md ? **? Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate. A10/A11 are recorded only for a limited product alpha capped at fewer than 25 individually allowlisted testers; #172 and #486 remain the model/public/commercial gates. §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
19 - CONTROL_PLANE_DURABLE_REBUILD.md ? **? Active tracked project (`DRAFT` owner-confirm-gated, created 2026-07-05):** the consolidated control-plane rebuild that unblocks tracker **DoD #1** after the first owner browser CUJ (2026-07-05) proved the deployed CP fails real usage (2.68 GB upload ? 4 GiB RAM-`/tmp` OOM ? restart ? ephemeral SQLite wiped ? detected rallies never ingested). One image rebuild + redeploy: **8 GiB CP · upload staging STREAMED to the bucket (never tmpfs) · bucket-grounded durable state (DB = cache, re-hydrate from `outputs/<md5>/`) · submit-time MD5 (instant re-runs + restart recovery) · honest poll ceilings · rev-00011 config pins baked into `gen_service_yaml.py` · SPA re-vendored (#153/#154/#158)**; bundles #482 progress + #483 golden-gated downsample when ready. Consolidates #471 #480 #482?#485. §7 tracker is the source of truth; **P9 (deploy) is owner-gated on spec confirmation**.
20 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.
21 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
22 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
23 - DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU*

enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

24

25 ## Practical / ongoing

26 - ALPHA_TESTER_ONBOARDING_FAQ.md ? **? Tester-facing onboarding + FAQ for the hosted alpha** (send-ready; owner-approved copy per the A10/A11 guardrails ? workflow claims only). Sign-in via email OTP at `api.khelsutra.guru` · what footage works · the upload?detect?pick?reel flow with honest timing expectations from the 2026-07-05 CUJ · resumable links · privacy pointer (`khelsutra.guru/privacy`) + delete-on-request · ground rules (personal invite, rights to footage, non-commercial) · known limits (4 GB cap, conservative quality gate being tuned, watermark).

27 - CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a §6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [`README.md`](../README.md); packaging roadmap ? [`PACKAGING_PLAN.md`](PACKAGING_PLAN.md).

28 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, $0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify_compute_path.py`) ? $0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? [`../deploy/cloudrun/README.md`](../deploy/cloudrun/README.md).

29 - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.

30 - DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:`/`F:`/Drive ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

31 - [`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: [`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).

32 - TRACKNET_WSL_SETUP.md

33 - Evaluation harness and calibration drivers (in `backend/eval/`)

34

35 ## Data layer (collection · annotation · golden sets ? **no ML**)

36 - [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.

37

38 ## Conventions & templates

39 - PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

40

41 ## Historical documents

42   Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
43
44   - **June-2026 test-hardening sweep** ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162) and **archived**; the former top-level pointer-stubs were removed 2026-07-02 (absorbed into the umbrella's *Prior audits*). Full records: [`archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md`](archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md), [`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md), [`archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md`](archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md).
45   - **CODE_CLEANUP_AUDIT (2026-06-24)** ? the prior deep engine audit (35 findings; Phase 1 complete, Phase 2 largely shipped, remaining = the `main.py` god-module split). **Absorbed into the umbrella** (see its *Prior audits (absorbed)*) and archived ? [`archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md`](archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md).
46   - **KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN** ? the five-repo audit execution board, **folded into the umbrella** as its *Remediation tracker (live)* and archived ? [`archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md`](archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md).
47   - DECISION_SNAPSHOT_REGRESSION.md ? **? Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.
48
49   - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ? [`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT_STEPS.md`).
50   - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ? [`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
51   - Superseded: per-video output layout ? [`archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (? `PER_VIDEO_WORKSPACE.md`).
52
53   See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.
54

24. user (2026-07-08T05:47:16.104Z)

13:[Omitted long matching line]
24:3b. **[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).** All waves (P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was archived to `docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus #301/#327/#332/#384.
153:   rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit

25. user (2026-07-08T05:47:17.720Z)

12:https://github.com/Khelsutra/badminton-highlight-indexer/issues/523
16:1. Engine PR for #523: add an opt-in compile rendition field (`original` default,
24:4. Link each PR back to #523 and add verification notes/artifacts there.

26. user (2026-07-08T05:47:27.200Z)

1   # Next steps (live) ? refreshed 2026-06-21
2

3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 > **? 2026-07-05 ? ACTIVE: the control-plane durable rebuild.** After the first owner browser CUJ
6 > blocked on the deployed CP (OOM/ephemeral-state ? DoD #1), the consolidated rebuild is specced,
7 > **owner-confirmed**, and executing: CONTROL_PLANE_DURABLE_REBUILD.md
8 > (§7 there is the source of truth ? P1?P8 PRs, then the P9 rebuild+redeploy+CUJ).
9
10 ## ? Where we are (2026-06-21)
11 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21; M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
12
13 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not** cleared for an external/commercial launch; we stay in **development**. This is the expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.
14
15 ## ? Prioritized pending ? all tracks
16 **P0 ? highest leverage / unblocks the most**
17 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
18 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
19 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
20 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.
21 3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the

`0.1.0-l4` bundle is n=2 plumbing-only).
22
23 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
24 3b. **[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).** All waves (P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was archived to `docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus #301/#327/#332/#384.
25 4. **[Claude] `COMPUTE_DECOUPLED_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked `consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `_KNOWN` + `_backend_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.
26 5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
27 6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
28 7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
29 8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.
30
31 **P2 ? later / gated / deferred**
32 9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
33 9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in `RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.
34 10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
35 11. **Multisport** (table-tennis first, then pickleball).
36 12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
37 13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
38 14. **Rename repo off "badminton"** (owner, BACKLOG).
39 15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
40 16. **PMF field validation** (owner-side, business track).

27. user (2026-07-08T05:48:09.912Z)

1 # DEPLOY_REPORT ? cloud-serving (M7) first real L4 run + GPU optimization sweep
2
3 > **Date:** 2026-07-04 · **Owner-authorized** GCP spend (D17, $300 envelope) · deleted every
4 > resource after (`[[gcp-vm-always-delete]]`; §5 residual audit = $0 of ours).
5 > **Video:** `MahadevpuraSingles.MP4` (owned golden, 13,418 frames). **Job:** Cloud Run L4
6 > (Ada sm_89, 24GB) · 8 vCPU / 32GB · `--no-gpu-zonal-redundancy` · scale-to-zero.
7 > **Mode:** fully-local (`skip_ai_handoff=true`) ? the number is the **owned WASB path only, zero Gemini**.
8
9 ## 1. M7 infra proven
10 - Worker image builds + pushes (`asia-southeast1/rally`); the M6 `CloudRunCompute` shim dispatches the
11 L4 Job and bucket-polls the done-marker; WASB weights via a GCS-FUSE read-only mount

(no weights in image ? WASB-C3); `intervals.csv` + `report.json` land in the bucket.
- Owned model runs on the L4: `native exec wasb_infer.py`, 13,418 trajectory points, 48 windows, 0 Gemini.

2. GPU optimization ? measured
Headline: the L4 is CPU-FEED-bound, not GPU-bound. GPU utilization sat **<1%** on every config ? a GPU that idle is *waiting*, not computing. So kernel/GPU levers do nothing; the wins (and the remaining headroom) are all on the CPU feed.

#	Config	Phase-2 wall	vs baseline	GPU util	Parity (rally windows)
0	cu113, batch 8 (baseline)	727.9s	?	<1%	reference
1	torch 2.2.2+cu118, batch 8	722.1s	?0.8% (noise)	<1%	intervals byte-identical; trajectory differs (6780 vs 6777 pts)
2	cu113, **batch 64 + prefetch 4**	**506.1s**	**?30.5% (1.44×)**	~1%	**byte-identical** ? (6777 pts)
3	cu113, batch 128 + prefetch 6	OOM	?	?	exceeds 24GB VRAM (`CUDA OOM, 9 GiB`)

Dispositions
- **? SHIP ? batch/prefetch tuning (config #2):** 30.5% parity-safe. Batch 64 is at the L4 VRAM ceiling (128 OOMs), so it's the max viable value, not arbitrary. ? PR `claude/wasb-gpu-batching`.
- **? DISCARD ? torch cu113?cu118 arch bump (config #1):** 0.8% = noise (JIT was never the wall, since the GPU was idle regardless), *and* it perturbs trajectory numerics (a byte-parity risk). Empirically dead for this workload.
- **? DON'T ? bigger GPU:** idle at <1% ? an A100/H100 would also idle. Pure waste (D8).
- **? OPT-IN ONLY (never default) ? FP16 / TF32 / cudnn.benchmark / NVDEC:** each changes detector numerics ? breaks the D5/D16 byte-parity guarantee (verified reasoning in the lever workflow).

3. Root cause & the remaining ~5× (scoped follow-up)
Streaming forces `DataLoader(num_workers=0)` (the in-memory `SequentialFrameStore` + `read_image` shim can't pickle to worker processes), so decode?PIL?ToTensor runs on **one core**, inline with the GPU; sliding windows re-transform each frame ~`frames_in`×. At batch 64 the GPU is *still* ~1% ? batching only removed the per-batch round-trip/Python overhead; the ~500s floor is the single-threaded feed. **The remaining 3?5× is a parallel-feed change**, not a config or a bigger machine:
- **Design:** partition the video into contiguous chunks, one DataLoader worker per chunk (each opens its own `VideoCapture`, seeks to its chunk start, reads forward ? sequential within a chunk, parallel across chunks); collate reorders. Parity-safe (same frames, same transform, same order).
- **Alternative (rejected for Cloud Run):** the existing disk-frames path already does `num_workers>0`, but materializing 13,418 PNGs into the RAM-backed `/tmp` (~27GB) OOMs the 32GB box ? which is *why* streaming exists. A real scratch disk (GCE VM, not Cloud Run) would unlock it.
- Only after the feed is parallel does a **higher-vCPU config** pay off (more cores to feed the GPU).

4. Cost ledger (D17)
- Builds: 4 × ~$0.16 (cu113 base reuse, cu118, batch) ? $0.5

```
53	- Runs: baseline + cu118 + batch64 + batch128(OOM, short) + probes ? ~$2?3 of L4 time
54	- Idle: **$0** (scale-to-zero; GPU bills only during execution)
55	- **Total well under the $300 / D17 envelope.**
```

28. user (2026-07-08T05:48:10.556Z)

```
backend\orchestration.py:53:_LOCAL_COMPUTE_LANE = threading.BoundedSemaphore(1)
backend\orchestration.py:918:            # Serialize the CPU/GPU-heavy in-process detection (see _LOCAL_COMPUTE_LANE). A remote
backend\orchestration.py:920:            with _LOCAL_COMPUTE_LANE:
backend\orchestration.py:1163:        # dispatch) doesn't run multiple compiles at once on the box (see _LOCAL_COMPUTE_LANE).
backend\orchestration.py:1164:        with _LOCAL_COMPUTE_LANE:
backend\api\job_worker.py:37:# orchestration._LOCAL_COMPUTE_LANE regardless of the worker count, so the single-box invariant holds
backend\api\job_worker.py:76:    stitch serialize via orchestration._LOCAL_COMPUTE_LANE regardless of the worker count ? so the
```

29. user (2026-07-08T05:48:18.438Z)

```
1	# Testing Strategy ? Compute-Decoupled Rally Serving
2
3	> **Companion to** [`README.md`](README.md) (§7 tracker is the source of truth). **Status:** DRAFT (2026-06-21).
4	> **Thesis:** the decoupling is only *real* if we can **prove the core is unaffected across profiles** ? for **$0, with no GPU/WSL/network** ? and quarantine the one thing that genuinely can't be byte-deterministic (cross-GPU detection) to an owner-side real-hardware layer. Tags: `[verified]` = building block already in the tree; `[design]` = to build.
5
6	---
7
8	## 0. The two guarantees we are testing
9	1. **Decoupling guarantee** ? the same `(input bytes + config)` produces the **same windows + the same stitched segment ranges** whether the core runs truly-local or on Cloud Run. *Provable for $0* because detection can be replaced by a deterministic `replay_csv` stub.
10	2. **Quality guarantee** ? real WASB detection on a real GPU is *good enough at the rally level*. **Not** byte-deterministic across GPU architectures (L4 vs 2070 differ sub-pixel ? 684/1195 frames, per the 2026-06-20 deploy report), so it is judged at the **window/rally level**, owner-side, on the golden corpus.
11
12	The whole strategy hangs on keeping these **separate**: CI gates #1 deterministically; the owner gates #2 on real hardware.
13
14	## 1. Test layers
15
16	### L1 ? Unit (pure core) `[verified]`
17	- **What:** `trajectory_to_action_windows` determinism; `_merge_overlapping`; `_watermark_filter` string build; `_parse_time`; **profile preset/precedence resolution** (new); `StorageRef.parse_uri` scheme dispatch.
18	- **How:** pytest, no GPU/net/WSL; same-input?same-output; `abs-tol 1e-6` on floats.
19	- **Modes:** all (pure fns are profile-independent).
20
21	### L2 ? Contract (seam ABCs) `[verified]`
22	- **What:** `DetectorRunner` ABC ? `run_predict` CSV schema + `healthcheck` non-raising ? across `Wasb`/`NativeWasb`/`Stub`; `StorageBackend` ABC roundtrip (`put ? exists ? stat ? list ? fetch ? delete`) across local/s3/gcs/gdrive; the new `DeviceContext` probe contract.
23	- **How:** stub runner for detection; **injected fake S3/GCS clients + gdrive=`tmp_path` mount** (`tests/test_storage.py`); assert the CSV columns are identical across runners.
24	- **Modes:** `cpu-mock` proves the contract; local/cloud use the same code paths.
25
26	### L3 ? Integration (GPU-free E2E) `[verified]`
27	- **What:** the full trajectory-hybrid pipeline (detect ? window ? persist ? stitch) on the stub detector; `worker cmd_infer` fetch ? segment ? put with fake storage; chunked-upload assemble.
28	- **How:** `RALLY_DETECTOR_IMPL=stub` (synth **and** `replay_csv`); storage fakes;
```

`tmp_path` workspace; assert `intervals.csv` + `report.json` + a reel are produced deterministically.

- **Modes:** `cpu-mock`, but it **exercises the cloud-serving code path** (fetch/put/dispatch) with no cloud.

L4 ? **Profile-PARITY (the load-bearing proof)** `[design]`

- **What:** the explicit decoupling guarantee ? run the **same `replay_csv` golden trajectory** through (a) the in-process **truly-local** path and (b) the **worker + fake-`gcs`** (cloud-serving-mock) path, and assert **byte-identical candidate windows + identical stitched segment ranges** (and, on a fixed clip, an identical reel via ffmpeg determinism).
- **How:** one test parametrized over `{profile=truly-local, profile=cloud-serving-mock}`, fed the same replay CSV + same windowing config; **diff the window list and segment ranges exactly.** Because detection is a deterministic replay here, **any divergence is necessarily a profile/seam leak** ? exactly what must never happen.
- **Modes:** truly-local vs cloud-serving (mocked). **This is the test that, if green, proves the whole design.**

L5 ? E2E real-GPU (owner-side) `[design]`

- **What:** real WASB detection on a real GPU (local 2070 ? truly-local; GCP L4 ? cloud-serving) end-to-end ? reel; cross-GPU parity checked at the **rally/window level** (NOT CSV bytes).
- **How:** owner runs the **golden corpus** on the GPU box and on Cloud Run; compare rally counts/windows; **smoke-test the reel** (`ffmpeg -f null -` full decode exit 0 + a 5-frame contact sheet visually confirming real rallies). Recorded in a **DEPLOY_REPORT** under `runlogs/`.
- **Modes:** truly-local + cloud-serving on **real hardware**.

2. Real-GPU vs mocked ? the split

- **Mocked is the CI default.** The stub detector (CPU, deterministic, **byte-exact via `replay_csv`**) + in-memory storage fakes prove **all plumbing + the profile-parity guarantee** with **$0** and no GPU/WSL/network.
- **Real-GPU is owner-side only** (the golden corpus + labelled videos live with the owner). It validates that `NativeWasbRunner` actually uses the GPU and that cross-GPU detection diffs are **sub-pixel** ? judged at the window level, never CSV bytes.
- **The contract:** CI gates the deterministic post-trajectory transforms (window + stitch) **byte-exact**; real-GPU gates that detection on the chosen GPU is **good enough at the rally level**. Honest-limits caveat (golden §6) baked in: **green CI proves plumbing/determinism, not field quality.**

3. CI gates (extend the existing lanes)

Today's lanes: `ruff`, `mypy` (lenient baseline), import-smoke (no GPU stack), full suite (offline, mocked, `--cov-fail-under`), golden-fixtures cross-OS (ubuntu/windows/macos). **Add:**

- **Profile-parity lane** ? the L4 test (truly-local vs cloud-mock identical windows/segments).
- **Storage-fakes roundtrip lane** ? L2 across all four backends.
- **Per-PR core-touching gate** ? any PR touching detect/window/stitch *callers* (esp. **M3**, the `output/` de-hardcode) must show: golden fixtures green on **all 3 OSes** AND `experiment.compare(CONTROL, profile-variant)` **F1-delta ? 0** (no-regression), AND **coverage not regressed** vs base.
- **No real GPU / net / WSL in CI** ? ever. Real validation is the owner's L5.

4. Profile-parity ? why it's airtight

The decoupling claim is proven by feeding one `replay_csv` golden trajectory through **both** the in-process path and the worker+fake-bucket path and asserting **byte-identical** windows + segment ranges. Detection is replaced by a deterministic replay, so **the only variable is the seam** (compute target + storage backend). If the outputs ever diverge, a profile leaked into the core ? the precise failure D1 forbids. Real cross-GPU detection diffs are **quarantined** to L5 and judged at the window level.

5. Sample runs / runlogs for posterity (the DECOUPLED convention)

Each real run writes a dated report under [`runlogs/`](runlogs/) (e.g. `DEPLOY_REPORT_<profile>_<date>.md`), mirroring `DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md`, capturing:

- resources provisioned + the **exact commands**;

- run results ? **rally counts, parity numbers**, `gpu_active_seconds` / `wall_seconds` / `bytes_out`;
- a **reel smoke-test** ? `ffmpeg -f null -` exit 0 + a 5-frame contact sheet visually confirming real rallies;
- gotchas (conda-ToS, WASB-C3, GPU stockouts, font fallback) + the **final teardown state** ($0 audit);
- so the next deploy is **one-shot**.

DoD: `truly-local` (M5) and `cloud-serving` (M7) each get **?1 committed sample run** here.

6. Coverage map ? what each milestone must prove before merge
Milestone	Must be green
M1 config	L1 preset/precedence; profile='' = today (no behaviour change)
M2 input_backend	L2 storage roundtrip (4 backends); L3 worker fetch with local=identity

30. assistant (2026-07-08T05:49:37.457Z)

I have everything needed. Compiling the structured code map for the DOCS + CONVENTIONS subsystem.

31. user (2026-07-08T05:51:52.379Z)

Structured output provided successfully